MolSanity: A Reliability Audit for Feature Attributions on Molecular Graph Neural Networks

Karthik Iyer[1,*] Nasser Sabar[1]

[1]Department of Computer Science and Information Technology, La Trobe University, Melbourne, Victoria 3086, Australia

*Correspondence: karthik.iyer@hotmail.com ORCID: 0009-0004-0593-1602

Abstract. Feature attributions are routinely used to justify molecular graph neural network (GNN) predictions to chemists, yet they are almost never audited for *reliability*: existing evaluation frameworks ask whether an explanation is faithful to the model, not whether the model's explanation can be trusted, and not where it stops being trustworthy under the scaffold shift that characterises real drug-discovery workflows. We present **MolSanity**, a reliability-audit framework that wraps canonical attribution implementations (Captum, PyTorch Geometric, RDKit) rather than proposing a new attributor, and scores every (dataset × backbone × attributor × split) cell on six axes: motif-native coherence, occlusion–attribution faithfulness, ground-truth localisation where node labels exist, cross-checkpoint stability, calibration linkage, and confidence/correctness regime stratification. Our central finding is that *faithfulness is not correctness*, and that the relationship between them collapses under shift. Holding dataset, backbone and attributor set fixed and changing only the split, all 3 faithfulness metrics we test (our occlusion measure, Fidelity+ and the GraphFramEx characterisation score) fail to recover the ground-truth-best attributor on MUTAG under a Bemis–Murcko scaffold split. The occlusion measure selects GuidedBP, whose agreement with the mutagenic nitro motif is GT AUROC 0.013 — near-perfectly *anti*-aligned, and sixth of the 7 attributors audited — while the ground truth ranks GNNExpl. best at 0.826 (median paired gap 0.967 AUROC, Wilcoxon $p < 0.001$, Benjamini–Hochberg $q < 0.001$ over the 18 selection tests). What separates the regimes is the rank correlation between faithfulness and correctness. Pooled over all 33 cells of the 3 molecular ground-truth arms (each an across-seed mean of 3 seeds), it moves from +0.009 in distribution ($p = 0.962$) to −0.353 under shift ($p = 0.044$). On MUTAG alone the same contrast is +0.143 to −0.643. We report that 2 of the 3 arms move in this direction and one does not, and say why in § rather than pooling the disagreement out of sight. We are equally explicit that faithfulness rankings mismatch the ground truth in-distribution as well (3 of 3 metrics on MUTAG), so the finding is a collapse in rank correlation rather than agreement followed by disagreement. Faithfulness itself does not fall under shift — across the same 33 cells it *rises*, from 0.028 to 0.132 ($p = 0.008$), while ground-truth localisation does not move (0.641 to 0.631, $p = 0.888$). The metric therefore moves in the reassuring direction in exactly the regime where its relationship to correctness breaks, and gives no warning. Using 26185 committed per-molecule records we further find that on confidently-wrong predictions ground-truth localisation degrades (0.790 to 0.674, $n = 257$) while their measured faithfulness *improves* (0.126 to 0.330) and their stability is no worse — both proxies point away from the failure — and that the calibration–reliability link attenuates from a per-cell median of 0.137 to 0.046 when cells are pooled, a Simpson's-paradox trap for anyone reporting a single aggregate number. Results are the committed `full.yaml` run: 474 of 474 cell-runs completed. Every number, figure and table regenerates from the committed artifacts.

Introduction

A medicinal chemist is shown a graph neural network's prediction that a candidate is hERG-active, together with a heat map over its atoms. The heat map concentrates on a piperidine nitrogen; the chemist finds that plausible, and the series is deprioritised. Nothing in that workflow ever established that the highlighted atoms are the atoms the model actually used, that they are the atoms that matter chemically, or that the highlighting behaves the same way on the next scaffold. Each of those is a separate question, and the answers come apart.

They come apart in a way that is measurable, and that gets worse in exactly the setting drug discovery operates in. An attribution can be *faithful*, in that the atoms it ranks highest really are the atoms whose removal moves the model's output most, and simultaneously *wrong*, in that those atoms are not the substructure that determines the property. Across the 114 committed cells in this study that carry a node-level ground truth, 13 are positively faithful to the model and yet *anti*-aligned with the true motif (ground-truth AUROC below the chance value of 0.5). A benchmark that scores only faithfulness rates all of them as fine.

In practice the problem takes the form of a selection problem. With no ground truth available, which is the situation on every real molecular dataset in this study, an attributor has to be chosen using a faithfulness or fidelity metric. On the two cell families that do have a ground truth and were audited on both splits, we can check whether that choice is the right one while changing *only* the split. Under a Bemis–Murcko scaffold split on MUTAG, all 3 ranking metrics (our occlusion measure, Fidelity+ and the GraphFramEx characterisation score alike) select GuidedBP, whose agreement with the mutagenic nitro motif is GT AUROC 0.013, while the ground truth ranks GNNExpl. best at 0.826. The median per-molecule gap is 0.967 AUROC (paired Wilcoxon $p < 0.001$, $n = 53$; Benjamini–Hochberg $q < 0.001$ across the 18 selection tests), and the rank correlation between faithfulness and correctness is *negative* (-0.643 to 0.036). In distribution on the same dataset, two of the three metrics instead select the ground-truth-best attributor and the rank correlation is $+0.143$.

None of this makes faithfulness metrics wrong. They answer a different question from the one a chemist is asking, and the gap between the two questions widens under distribution shift. Molecular property prediction is deployed under scaffold shift almost by construction, since the point of a screening model is to score series the training set does not contain, and scaffold-split evaluation is now standard practice for the *predictions* themselves [26, 27]. The evaluation protocol for the *explanations* has not followed: it does not stratify by that shift at all.

Contributions. This paper presents **MolSanity**, a reliability-audit framework for feature attributions on molecular GNNs. Rather than propose another attribution method, it wraps the canonical implementations (Captum [17], PyTorch Geometric [18], RDKit [19]) and audits them. Specifically:

- We define a six-axis audit over a motif-native (RDKit) decomposition: coherence, occlusion–attribution faithfulness, ground-truth localisation, cross-checkpoint stability, calibration linkage and confidence/correctness regime stratification, with field-standard Fidelity±, sparsity, the GraphFramEx characterisation score [10] and the PyG/DIG unfaithfulness metric [11] computed on the *same* molecules for direct comparability (The MolSanity audit).
- We show that a faithfulness-only ranking picks the wrong attributor under scaffold shift, in a design where only the split varies, so the effect is not confounded with a change of dataset (A fidelity-only ranking picks the wrong attributor under shift).
- We separate what shift breaks from what it leaves alone. The *backbone* ordering does not survive the change of split ($\rho = 0.100$ over 5 backbones), so which architecture localises best in distribution says little about which does under shift. The *attributor* ordering largely does survive ($\rho = 0.821$ on MUTAG, 0.964 on MolMotif), and the level of faithfulness does not shift systematically either (across the 79 cells audited on both splits the median change is 0.048). What breaks is neither of those but the *relationship* between faithfulness and correctness — which is precisely the quantity a faithfulness-only benchmark cannot see.
- Using 26185 per-molecule audit records, we stratify reliability by confidence/correctness regime and test the calibration-linkage hypothesis directly. It holds per cell (median $\rho = 0.137$, 69 cells significantly positive vs. 26 negative) but *attenuates to* 0.046 when cells are pooled, a Simpson's-paradox trap for anyone reporting a single aggregate number.
- We report the negatives with the positives: a regression faithfulness operator we mis-specified and whose numbers we withdraw, ground-truth arms whose scaffold split is degenerate and are therefore excluded from the shift contrast rather than counted toward it, an exactly-labelled molecular arm too saturated to adjudicate the central claim, and the 44 cells where the correctness axis simply does not exist and is marked as such rather than scored.

Related work

Attribution methods. The attributors audited here are the standard ones. The gradient family (Saliency [2], Input×Gradient, Guided Backpropagation [3] and Integrated Gradients [1]) is model-agnostic and cheap. Graph-specific methods learn a mask over the input: GNNExplainer [4] optimises a soft edge/feature mask per instance, PGExplainer [5] amortises that into a parametric mask network, and SubgraphX [6] searches connected subgraphs with Monte-Carlo tree search under a Shapley objective. Graph analogues of CAM/Grad-CAM were introduced by Pope et al. [7]; Yuan et al. [8] survey the space. In every case we call the published implementation rather than reimplementing it.

Evaluating explanations. Two evaluation traditions exist. *Faithfulness* measures whether the explanation tracks the model: Fidelity± and sparsity [8], the GraphFramEx characterisation score that combines them [10], and the unfaithfulness metric shipped with DIG/PyG [11]. *Correctness* measures whether the explanation recovers a known rationale, which requires ground-truth substructures, typically synthetic [9]. The tension between the two is known: Faber et al. [13] argue that comparing against ground truth is misleading when the trained model does not use that rationale, and Sanchez-Lengeling et al. [14] built the first systematic attribution benchmark for molecular GNNs on exactly this basis. Outside graphs, sanity checks [15] and retraining-based benchmarks [16] made a parallel case that plausible-looking attributions can be uninformative. MolSanity's position is that neither axis subsumes the other, and that what needs measuring is the *joint* behaviour of the two under distribution shift.

Frameworks, and the explicit delta. Four lines of work are close enough that a citation is not a sufficient answer. Table 1 scores all of them on the same capabilities; what follows states what each row of that table means.

Sanchez-Lengeling et al. [14] is the nearest neighbour and the earliest. It built the first systematic attribution benchmark for molecular GNNs on tasks whose ground-truth substructure is known by construction, establishing that attribution methods should be scored against a known rationale rather than against how plausible a chemist finds the highlighting. MolSanity inherits that premise. What it adds is the faithfulness axis measured on the *same* molecules and the joint behaviour of the two. A benchmark that reports correctness alone cannot show that a faithfulness-based choice of attributor is the wrong choice, because it never makes that choice. The selection-consistency test of The selection-consistency test makes it, and its outcome changes with the split.

MolFaith [12] is the closest work on the faithfulness axis and is larger than this study on that axis: eight attribution methods, two molecular representations and five architectures over roughly 14,000 molecules, against the 26185 per-molecule records here. We make no claim to broader faithfulness coverage. Two of its findings bear directly on ours. First, it reports that some methods are inherently more faithful than others largely independently of architecture. We find the same stability from the other direction: faithfulness does not shift systematically across the 79 cells audited on both splits (median change 0.048). Two independent studies at different scales agreeing that faithfulness is a comparatively stable property of the method is the premise our result then acts on. The correctness ordering is also fairly stable across the split ($\rho = 0.821$ on MUTAG, 0.964 on MolMotif). Both rankings are stable; what changes is how they sit relative to each other, and under shift they sit at odds. That is the distinction we would press: *stability of a proxy is not validity of a proxy*. A practitioner with no rationale labels who selects a method by faithfulness inherits a reproducible answer, not a correct one. Second, MolFaith reports that faithfulness varies markedly per molecule; our regime stratification locates part of that variation, since ground-truth localisation degrades precisely on confidently-wrong predictions while faithfulness on those same molecules *improves*.

GraphXAI [9] and *GraphFramEx* [10] are general-graph frameworks: GraphXAI supplies generators with exact node ground truth and an evaluation library, GraphFramEx systematises fidelity-style evaluation across explainers and tasks. Both are broader than MolSanity across graph domains and narrower within chemistry. MolSanity scores chemically meaningful units (SSSR rings, Murcko scaffold, BRICS fragments) rather than individual atoms, and stratifies by the split that governs molecular deployment. *DIG* [11] is a library rather than a protocol; we wrap its unfaithfulness metric, and its SubgraphX implementation is the coverage gap recorded in Limitations and threats to validity.

What is new here, and what is not. Ground-truth localisation, fidelity metrics, motif decompositions and scaffold splits all exist already. Two rows of Table 1 we do not find in any of the four: cross-checkpoint stability and calibration linkage. The contribution we would defend is neither of those rows alone, but the result they were built to support: that the two evaluation traditions above come apart under the shift molecular models are deployed into — far enough that the faithfulness ranking and the correctness ranking reverse relative to one another — and that no metric on the faithfulness side signals it.

The MolSanity audit

Notation

A molecule is a graph $G = (V, E)$ with node features $X \in \mathbb{R}^{|V|\times d}$ and edge features $Z \in \mathbb{R}^{|E|\times d'}$. A trained model f_θ maps G to logits $f_\theta(G) \in \mathbb{R}^C$ (classification) or a scalar (regression); t indexes the explained output. An *attributor* is a map $A : (f_\theta, G, t) \mapsto a \in \mathbb{R}^{|V|}$ assigning a real score to every atom.

RDKit decomposes G into a motif cover $\mathcal{M}(G) = \{M_1, \dots, M_K\}$, $M_k \subseteq V$, from SSSR rings, the Bemis–Murcko scaffold [31] and BRICS-style fragments. Writing $a_v^+ = \max(a_v, 0)$, the motif-level score is

$$s_k = \sum_{v\in M_k} a_v^+, \qquad k = 1, \dots, K. \tag{1}$$

Working at motif rather than atom granularity is what makes the audit chemically legible: a chemist reasons about a nitro group, not about atom 14.

For $S \subseteq V$ let $G\backslash S$ denote G with the features of S zeroed through the model's node-mask multiplier, $X' = X \odot$

Table 1: Where the audit sits relative to existing evaluation frameworks. ✓ = core capability, ∼ = partial or possible but not central, ✗ = not a focus. The contribution is the combination, not any single row: ground-truth validation and faithfulness both exist elsewhere, but not jointly with cross-checkpoint stability, calibration linkage and shift stratification over a motif-native decomposition.

Capability	GraphXAI	GraphFramEx	DIG	MolFaith	MolSanity
Ground-truth explanation benchmarks	✓ (synthetic)	∼	✓	✗	✓ (synthetic + motif-proxy)
Faithfulness / fidelity metrics	∼	✓	✓	✓	✓ (reproduced for comparability)
Molecular-motif-native audit (RDKit)	✗	✗	∼	∼	✓ (SSSR + Murcko + BRICS)
Occlusion–attribution agreement	∼	✓	∼	✓	✓
Cross-checkpoint stability	✗	✗	✗	✗	✓
Calibration linkage	✗	✗	✗	✗	✓
Scaffold-shift regime stratification	✗	∼	✗	✗	✓ (the core differentiator)
Multiple GNN backbones (agnostic)	∼	✓	✓	∼	✓ (GINE/GCN/GAT/MPNN/AttentiveFP)
Paired statistics (Wilcoxon, bootstrap CI)	∼	∼	∼	∼	✓
Across-seed variance reported per cell	✗	✗	✗	✗	✓ (3 seeds, 'SEED_VARIANCE.md')
Wraps canonical implementations (no re-impl)	✓	✓	✓	✓	✓ (Captum/PyG/RDKit)

$(\mathbf{1} - \mathbb{1}_S)$, leaving the topology intact.

The five audit statistics

(i) Faithfulness. Occlusion of motif M_k changes the explained output by

$$\Delta_k \;=\; f_{\theta,t}(G) \;-\; f_{\theta,t}(G \setminus M_k). \tag{2}$$

An attribution is faithful when it ranks motifs as occlusion does, measured by the Spearman rank correlation over the K motifs of one molecule,

$$\rho_{\text{occ}}(G) \;=\; \text{Spearman}\big((s_k)_{k\le K},\, (\Delta_k)_{k\le K}\big) \in [-1,1]. \tag{3}$$

$\rho_{\text{occ}} = 1$ means the attributor recovers the causal ordering exactly; $\rho_{\text{occ}} = 0$ means it is uninformative about it; and $\rho_{\text{occ}} < 0$ means it *inverts* it, which is strictly worse than an uninformative explanation. Undefined cases ($K < 2$, or zero variance in either argument) are recorded as missing, never as 0.

We also reproduce the field-standard fidelity pair [8]. With τ the 80th percentile of a^+ and $S_\tau = \{v : a_v^+ \ge \tau\}$ the salient set, and $p_t(\cdot)$ the softmax probability of the explained class,

$$\text{Fid}^+ = p_t(G) - p_t(G \setminus S_\tau), \tag{4}$$
$$\text{Fid}^- = p_t(G) - p_t(G \setminus \bar{S_\tau}), \tag{5}$$

higher Fid^+ and lower Fid^- being better, with sparsity $1 - |S_\tau|/|V|$. The GraphFramEx characterisation score [10] combines them as the weighted harmonic mean

$$\text{char} \;=\; \frac{(w_+ + w_-)\,\text{Fid}^+\,(1-\text{Fid}^-)}{w_-\text{Fid}^+ + w_+(1-\text{Fid}^-)}, \qquad w_+ = w_- = \tfrac{1}{2}, \tag{6}$$

undefined outside $\text{Fid}^\pm \in [0,1]$. That constraint matters for the regression cells, where Δ is taken in output space and $\text{Fid}^\pm$ leaves that range.

(ii) Correctness. Where a node-level ground-truth mask $g \in \{0,1\}^{|V|}$ exists, correctness is the ability of a to separate the true substructure from the rest, scored by the area under the ROC curve of the min–max normalised attribution against g,

$$\begin{aligned}\text{GT-AUROC}(G) \;=\; &\Pr[\tilde{a}_u > \tilde{a}_v \,|\, g_u{=}1,\, g_v{=}0] \\ &+ \tfrac{1}{2}\Pr[\tilde{a}_u = \tilde{a}_v],\end{aligned} \tag{7}$$

with u, v drawn uniformly from the positive and negative atoms. Chance is $\frac{1}{2}$; values below $\frac{1}{2}$ mean the attribution is *anti-aligned*, ranking the true motif systematically *below* the rest. When g is degenerate ($\sum_v g_v \in \{0, |V|\}$) the quantity is undefined and reported as such.

(iii) Coherence. Concentration and connectedness of the attribution mass: the Gini coefficient of a^+, the fraction of mass in the top 20% of atoms, the largest-connected-component fraction of S_τ in the induced subgraph, and the motif top-1 share $\max_k s_k / \sum_k s_k$. These are descriptive; Cross-task reliability shows they carry little reliability signal.

(iv) Stability. With θ^{early} an intermediate checkpoint of the same training run and $s^{\text{early}}, s^{\text{final}}$ the corresponding motif scores (1),

$$\text{stab}(G) \;=\; \text{Spearman}\,(s^{\text{early}},\, s^{\text{final}}). \tag{8}$$

An explanation that reorders motifs between two checkpoints of one run is a weak basis for a chemistry decision.

(v) Regime and calibration. Logits are temperature-scaled on the validation fold [32]; let $c(G) = \max_j p_j(G)$

be the calibrated confidence and $y(G) \in \{0,1\}$ the correctness of the prediction. With $\tau_c = 0.8$, each molecule is assigned

$$R(G) = \begin{cases} \text{conf-correct} & c \geq \tau_c,\ y = 1, \\ \text{conf-error} & c \geq \tau_c,\ y = 0, \\ \text{borderline} & c < \tau_c. \end{cases} \quad (9)$$

Calibration linkage is the within-cell rank correlation between per-molecule calibration quality and reliability,

$$\rho_{\text{cal}} \;=\; \text{Spearman}\,(1 - |c - y|,\ \rho_{\text{occ}}), \quad (10)$$

positive values supporting the hypothesis that better-calibrated predictions carry more trustworthy attributions.

The selection-consistency test

The audit's central question is operational rather than descriptive. A practitioner with no ground truth must choose an attributor using a faithfulness statistic; we ask whether that choice agrees with the one the ground truth would make.

Fix a *cell* $c = (\mathcal{D}, f_\theta, \pi)$, meaning a dataset, a trained backbone and a split, together with a set $\mathcal{A}$ of attributors, each audited on the same molecule sample $\mathcal{G}_c$. For attributor A write $\bar{m}(A) = |\mathcal{G}_c|^{-1} \sum_{G \in \mathcal{G}_c} m_A(G)$ for the cell mean of statistic m. Define the ground-truth-optimal and the faithfulness-optimal attributor

$$A^\star = \arg\max_{A \in \mathcal{A}} \overline{\text{GT}}(A), \qquad \hat{A}_m = \arg\max_{A \in \mathcal{A}} \bar{m}(A), \quad (11)$$

for each faithfulness statistic $m \in \{\rho_{\text{occ}}, \text{Fid}^+, \text{char}\}$. Two quantities follow. The *selection gap* is the paired per-molecule difference in correctness between the two choices,

$$\delta_m(G) = \text{GT}_{A^\star}(G) - \text{GT}_{\hat{A}_m}(G), \qquad G \in \mathcal{G}_c, \quad (12)$$

tested against H_0 : median $\delta_m = 0$ with a two-sided Wilcoxon signed-rank test on the molecules both attributors audited. The *selection-consistency coefficient* is the rank correlation between the two orderings over $\mathcal{A}$,

$$\rho_m^{\text{sel}} \;=\; \text{Spearman}_{A \in \mathcal{A}}\,(\bar{m}(A),\ \overline{\text{GT}}(A)) \in [-1, 1]. \quad (13)$$

$\rho^{\text{sel}} \approx 1$ means faithfulness is a sound surrogate for correctness in that cell; $\rho^{\text{sel}} \approx 0$ means it is uninformative; $\rho^{\text{sel}} < 0$ means it is actively misleading, in that ranking by faithfulness does worse than ranking at random. Because $\mathcal{D}$, f_θ and $\mathcal{A}$ are held fixed and only π varies between the two arms of each comparison, any change in ρ^{sel} is attributable to the split rather than to a change of dataset or model.

Estimation and inference

Cell means are reported with 95% percentile bootstrap intervals over molecules (2000 resamples, fixed seed). Attributor comparisons within a cell are paired on molecule identity and tested with the two-sided Wilcoxon signed-rank test; correlations are Spearman throughout, as all statistics here are ordinal in nature and several are heavy-tailed. Sample sizes are the audited molecules per cell (20–120), which is small enough that we treat individual p-values as descriptive and report them unadjusted for multiplicity, leaning on replication across cells rather than on any single test. Where a statistic is undefined for a molecule it is dropped from that statistic's mean only, and the retained n is reported alongside.

Implementation

The pipeline is staged and idempotent: each stage writes a marker keyed to a content hash of its configuration, every cell is isolated so one failure cannot abort a sweep, and trained checkpoints are content-addressed so an audit reproduces without retraining. Seeds, library versions, git revision and hardware are recorded per run. Attributors are the canonical implementations [17, 18], wrapped behind one interface rather than reimplemented.

Experimental setup

Datasets. Tier-1 provides the ground-truth anchor: **SynthMotifs** and its larger twin **SynthMotifsXL**, generated offline (Barabási–Albert base plus house/cycle motifs) with *exact* node ground truth, and **MUTAG** [29, 30] with the nitro proxy. Tier-2 is MoleculeNet [27]: **BBBP** and **BACE** (classification), **ESOL**, **FreeSolv** and **Lipophilicity** (regression). Tier-3 covers chemistry-meets-medicine endpoints: **ClinTox**, **SIDER** and **Tox21** as single-task binary views of the MoleculeNet panels, and **DILI** (drug-induced liver injury) and **hERG** (cardiotoxicity) via the Therapeutics Data Commons [28]. TDC sets are featurised with the same PyG encoding as MoleculeNet so the backbones are drop-in. BA-2Motifs and ShapeGGen [9] are attempted and gracefully skipped: the former's download returns HTTP 403 in this environment, the latter's install depends on the pre-2.5 PyG compiled extensions. Credential-gated resources are out of scope and never worked around.

Backbones. GINE [20, 21], GCN [22], GAT [23], MPNN [24] and AttentiveFP [25], sharing a pooling head, calibration and splits.

Attributors. Integrated Gradients, Saliency, Guided Backpropagation and Input×Gradient through Captum [17]; GNNExplainer and PGExplainer through PyTorch Geometric [18]. SubgraphX [6] requires DIG, which installs but cannot import in this environment (no `torch_sparse` wheel at the pinned torch version), so it is left blocked rather than approximated. PGExplainer is classification-only by construction: its mask network trains against class logits, so it is absent from the regression cells.

Splits. The primary split is a deterministic, class-aware Bemis–Murcko scaffold split [31], the protocol used to evaluate molecular property models under realistic generalisation [26] (whole scaffold groups move together, so there is no leakage, but every fold is guaranteed to contain every class), with a random split as the in-distribution reference. Imbalanced classification uses inverse-frequency class weighting and AUC-based model selection. Every comparison in this paper that is labelled "shift" versus "in-distribution" is these two splits on the same dataset.

Scale of the study. The sweep trains every (dataset, backbone) pair on both splits and audits every (dataset, backbone, attributor, split) combination, giving 474 audited combinations and 26185 per-molecule audit records in total. Audited samples are 20–120 molecules per combination, drawn from the held-out test fold. Exact software versions, hardware, seed and the per-combination outcome ledger are recorded with the released artifacts (Data and code availability).

Coverage. Of the 474 planned combinations, 474 completed and 0 could not be attempted because a dependency (ShapeGGen) is unavailable in this environment (Table 2).

A further 0 rows in the results matrix come from an earlier, reduced-budget run on different hardware: they cover attributor sweeps on the Tier-2/3 molecular sets that the primary configuration does not schedule, plus the SynthMotifsXL arm. These are genuine measurements, but they are not comparable in budget to the rest, so they are marked [c] throughout, excluded from every aggregate computed over per-molecule records, and excluded entirely from the analysis in A fidelity-only ranking picks the wrong attributor under shift. In total the matrix holds 146 classification and 12 regression rows, of which 114 carry a node-level ground truth.

Results

Table 2: The run ledger. Outcome of every cell the `full.yaml` sweep attempted, as recorded in `results/PROGRESS.md`. *carried* counts cells whose row still appears in the results matrix from an earlier reduced-budget CPU run because this run's attempt failed. Those rows are marked [c] throughout this paper and are never mixed into a this-run aggregate.

dataset	done	failed	skipped	carried
BA-2Motifs	36	0	0	0
BACE	12	0	0	0
BBBP	42	0	0	0
ClinTox	12	0	0	0
DILI	6	0	0	0
ESOL	24	0	0	0
FreeSolv	6	0	0	0
Lipophilicity	6	0	0	0
MUTAG	66	0	0	0
MolMotif	66	0	0	0
MolMotifHard	66	0	0	0
SIDER	12	0	0	0
ShapeGGen	42	0	0	0
SynthMotifs	66	0	0	0
Tox21	6	0	0	0
hERG	6	0	0	0
total	**474**	**0**	**0**	**0**

Failure reasons. .

Faithfulness is not correctness

Table 6 lists every committed cell in which attribution *correctness* can be measured at all, and Figure 1 plots the joint distribution. Of the 114 cells with both metrics defined, 27 are anti-aligned with the ground truth (GT AUROC < 0.5) and 13 of those are simultaneously positively faithful to the model. That combination is the failure mode a faithfulness-only audit cannot see, because from the model's point of view nothing is wrong.

The clearest molecular instance is the MUTAG scaffold block of Table 6, where dataset, backbone and split are fixed and only the attributor varies. Guided Backpropagation, Saliency and Input×Gradient are the three *most* faithful attributors on that cell ($\rho = 0.616, 0.406, 0.499$) and the three *worst* at recovering the nitro motif (GT AUROC 0.013, 0.002, 0.079). GNNExplainer, which the ground truth ranks best at 0.826, is less faithful (0.380) than all three. On this cell the two axes are not merely uncorrelated; they are inverted.

A fidelity-only ranking picks the wrong attributor under shift

If faithfulness and correctness dissociate, then ranking attributors by a faithfulness metric, which is what an evaluation framework must do when no ground truth

Table 3: Molecular regression cells. Occlusion faithfulness is computed in output space (predicted-value shift) rather than probability space. In 0 of the 12 committed regression cells the attribution–occlusion agreement is *negative*: the atoms an attributor ranks highest are not the atoms whose removal moves the prediction most. ESOL·GAT, ESOL·GCN, ESOL·GINE, FreeSolv·GINE, Lipophilicity·GINE run positive. Note the Fidelity$\pm$ columns leave the $[-1, 1]$ range a probability-space fidelity would occupy (up to 7.50), which is why we report but do not interpret their sign. Bold marks the highest occlusion ρ per dataset (emphasis only). PGExplainer is classification-only and therefore absent by construction.

dataset	backbone	attributor	split	n	RMSE	MAE	R^2	motif top-1	occ. ρ	occ. top-1	Fid+	stab.
ESOL	GAT	IG	random	200	0.764	0.568	0.867	0.831	0.803	0.933	3.11	0.947
ESOL	GAT	IG	scaffold	200	1.080	0.806	0.740	0.864	**0.855**	0.972	7.50	0.955
ESOL	GCN	IG	random	200	0.947	0.737	0.797	0.854	0.423	0.650	−0.84	0.965
ESOL	GCN	IG	scaffold	200	1.071	0.867	0.747	0.887	0.437	0.767	−0.77	0.867
ESOL	GINE	GNNExpl.	random	200	0.833	0.629	0.842	0.849	0.634	0.832	−1.19	0.958
ESOL	GINE	GNNExpl.	scaffold	200	1.005	0.768	0.778	0.879	0.464	0.817	−0.52	0.873
ESOL	GINE	IG	random	200	0.833	0.629	0.842	0.865	0.683	0.830	−1.46	0.932
ESOL	GINE	IG	scaffold	200	0.988	0.743	0.785	0.897	0.409	0.753	−0.65	0.909
FreeSolv	GINE	IG	random	193	1.483	1.065	0.830	0.865	0.393	0.772	−0.69	0.862
FreeSolv	GINE	IG	scaffold	193	1.336	0.999	0.811	0.865	**0.421**	0.898	−0.61	0.703
Lipophilicity	GINE	IG	random	200	0.773	0.591	0.569	0.786	**0.556**	0.737	0.32	0.749
Lipophilicity	GINE	IG	scaffold	200	0.920	0.705	0.424	0.826	0.494	0.712	0.51	0.762

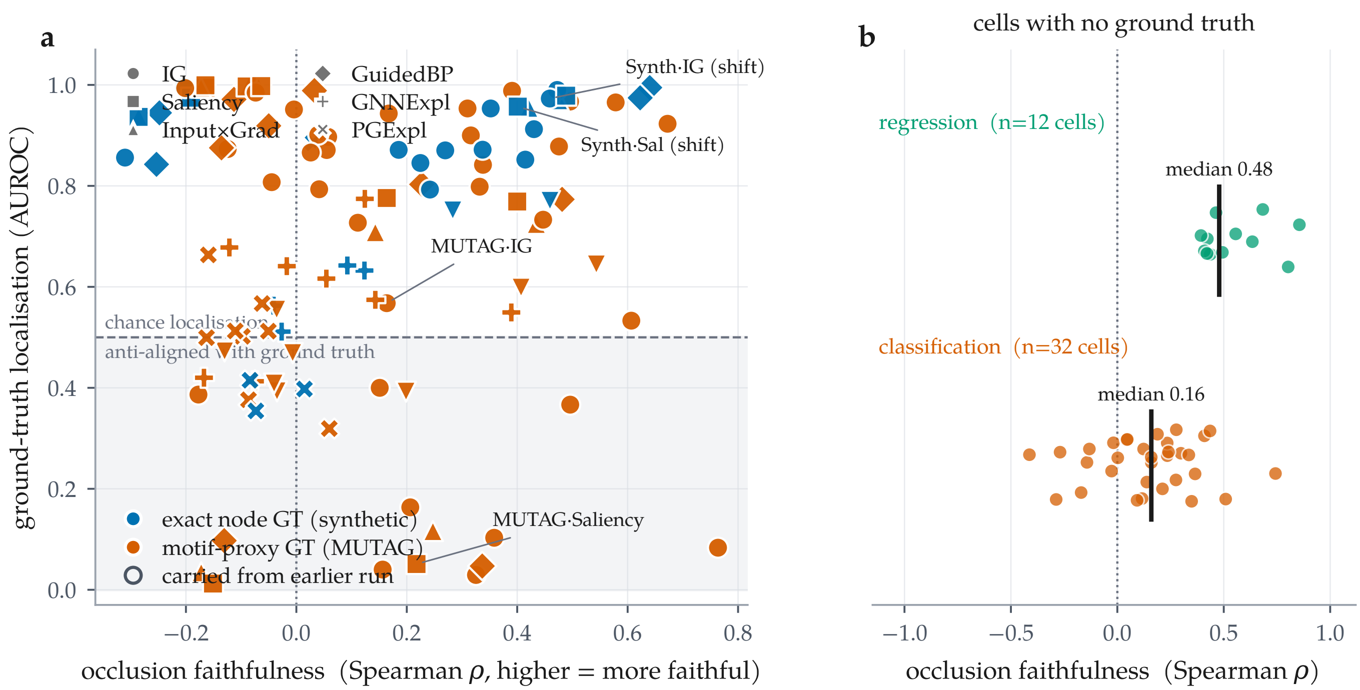


Figure 1: The two axes come apart, and for most cells only one of them exists. (**a**) Every committed cell carrying a node-level ground truth ($n = 114$), plotted as occlusion faithfulness against ground-truth localisation; colour distinguishes exact synthetic ground truth from the MUTAG motif proxy, marker distinguishes attributor, and open markers are rows carried from an earlier run. The shaded band is GT AUROC < 0.5: attributions there are anti-aligned with the true motif. 13 cells sit in the lower-right region, positively faithful to the model and anti-aligned with the truth. (**b**) The remaining 44 combinations, on real molecular datasets, have no published per-atom labels: only the horizontal axis is measurable for them.

is available, can select the wrong method. Two cell families let us test this cleanly: each has a node-level ground truth, all 7 attributors audited on the same molecules, and both splits from this run, so the in-distribution and shift arms differ *only* in the split (Table 5, Figure 3).

MUTAG, under shift. The ground truth ranks GNNExpl. best (0.826). All 3 metrics instead rank Guid-

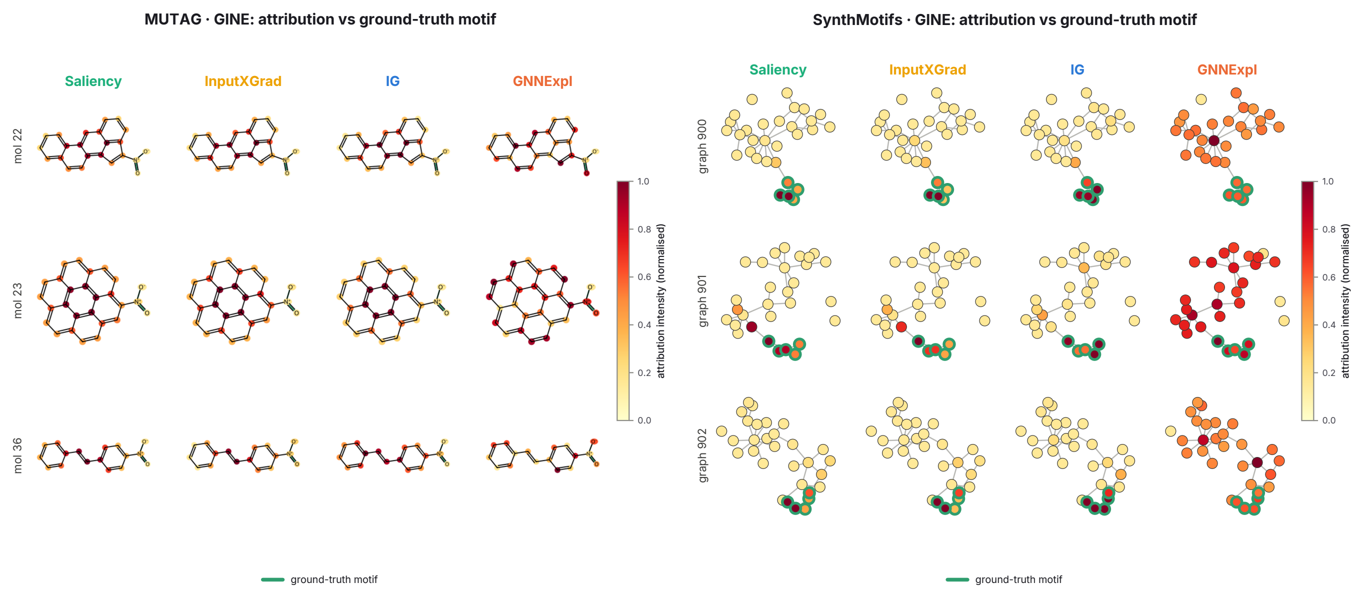


Figure 2: What the disagreement looks like on the structures themselves. Attribution intensity overlaid on the audited graphs for four attributors on the same molecules, rendered by the audit run: RDKit skeletal structures for MUTAG (left) and node-link diagrams for SynthMotifs (right), with the ground-truth motif outlined in green. The quantitative gap of Table 5 is visible here as a qualitative one: on the synthetic graphs the gradient attributors concentrate on the outlined motif while GNNExplainer spreads mass across the Barabási–Albert base, and on MUTAG the gradient methods saturate the fused aromatic system rather than the nitro groups that carry the mutagenicity label. SynthMotifs appears here as a structural illustration only: it is not a molecular dataset, so it has no Bemis–Murcko scaffold and takes no part in the shift contrast (§). These panels are the pipeline's own output, reproduced unmodified.

edBP first, at 0.013, the worst of the 7. On paired per-molecule GT AUROC over the 53 shared molecules the median gap is 0.967 with $p < 0.001$, and $q < 0.001$ after controlling the false discovery rate over all 18 selection tests, so this contrast is not an artefact of testing many of them. The faithfulness–truth rank correlation is negative for every metric: -0.643 for occlusion ρ, -0.107 for Fidelity+ and 0.036 for characterisation. A framework ranking on faithfulness here fails to find the best attributor and systematically prefers the worst.

MUTAG, in distribution. The same experiment on the random split behaves differently. Occlusion ρ and Fidelity+ both select GuidedBP, which is also the ground-truth-best, and their rank correlation with truth is $+0.143$. The characterisation score is the exception, selecting SubgraphX at 0.348 ($p < 0.001$)with rank correlation 0.821. The sign flip between the two arms is large ($+0.143$ to -0.643 for occlusion ρ), though this does not make in-distribution evaluation safe, only much less dangerous.

The pooled result, and the arm that dissents. Any single arm gives a rank correlation over seven attributors, which is a weak instrument. Pooled over all 33 cells of the 3 molecular ground-truth arms, the faithfulness–correctness correlation moves from $+0.009$ in distribution ($p = 0.962$) to -0.353 under scaffold shift ($p = 0.044$), while faithfulness itself rises (0.028 to 0.132, $p = 0.008$) and ground-truth localisation does not move ($p = 0.888$).

Per arm, 2 of 3 move in that direction: MUTAG strongly ($+0.36$ to -0.75) and MolMotifHard weakly (-0.29 to -0.57). MolMotif moves the other way (-0.64 to -0.07), and we report that rather than let the pool absorb it. The structural difference is visible in the ranking itself. On MUTAG no attributor exceeds GT AUROC 0.9 and the seven are spread across the range; on both MolMotif arms four of seven sit above 0.9, within a few hundredths of each other, so the rank correlation is carried by whether three poor attributors happen to order correctly against four that are effectively tied. A Spearman coefficient over such a distribution is not a sensitive instrument, and we would not draw a conclusion from any of the three arms on its own.

MolMotifHard exists because of this. MolMotif labels on halogen-aromatic presence, which a message-passing model can satisfy by detecting a rare *element* rather than an arrangement; MolMotifHard labels on a carboxylic acid, built from carbon and oxygen alone, so no single-element shortcut is available. It widens the usable range (GT AUROC spanning 0.38–1.00 against 0.47–1.00) without removing the bunching at the top, which is why it helps the pooled estimate without settling the question by itself.

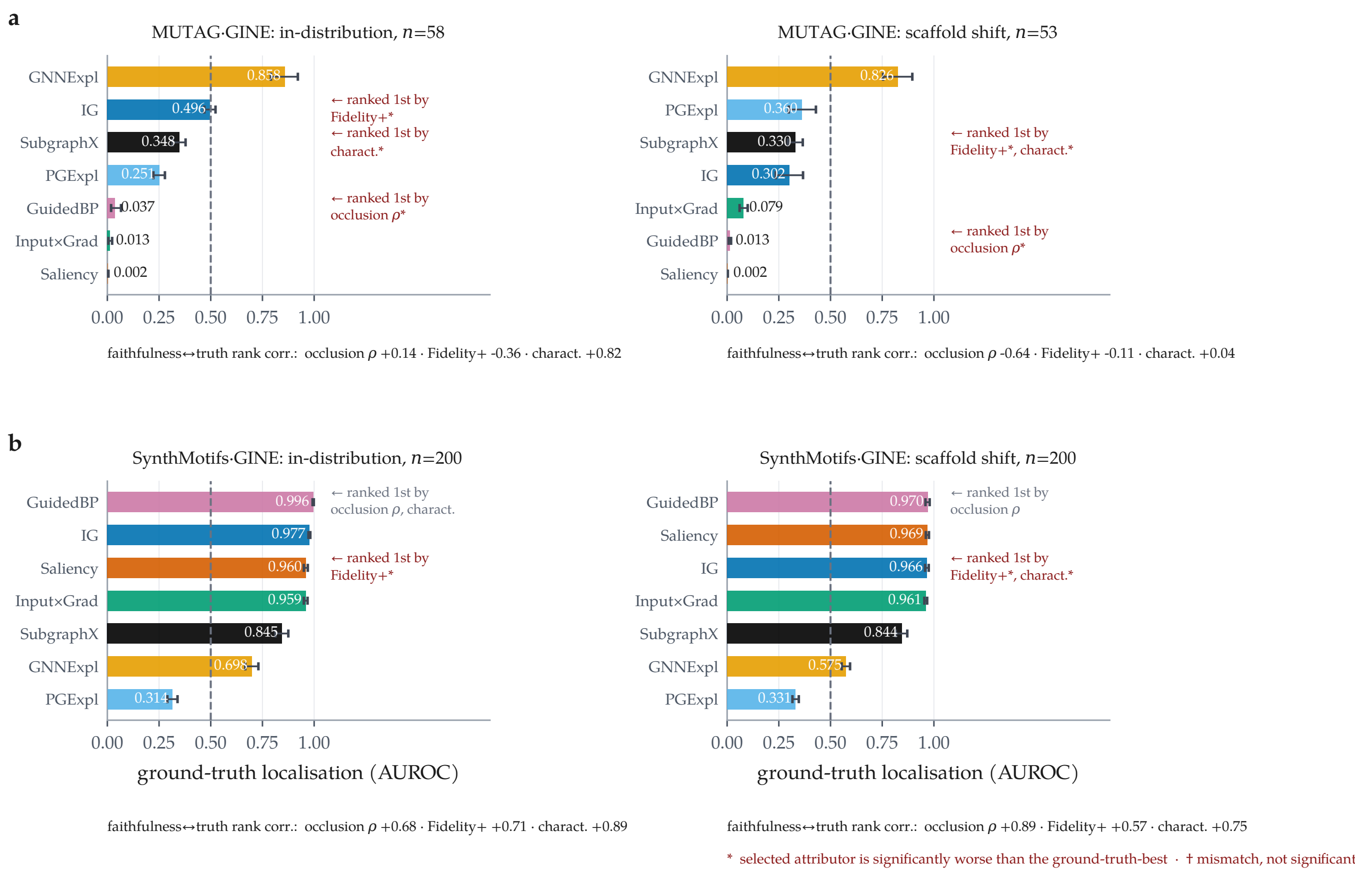


Figure 3: Under scaffold shift a faithfulness-only ranking selects the wrong attributor; in distribution it mostly does not. Ground-truth localisation per attributor, with bootstrap 95% confidence intervals over molecules. Within each row the dataset, backbone and attributor set are fixed and only the split changes. (**a**) MUTAG (motif-proxy ground truth): all 3 metrics mismatch the ground-truth-best attributor in both regimes, but only under shift does the occlusion measure select GuidedBP, whose localisation (0.013) is anti-aligned with the motif; significant picks are marked *, picks statistically indistinguishable from the best are marked †. (**b**) MolMotif (real molecules, exact node ground truth): the selection failure replicates, but the top attributors sit at GT AUROC $\geq$ 0.970, so the ranking among them is noise around a ceiling and this arm cannot adjudicate the regime contrast. Rank correlations between each faithfulness metric and ground truth are given below each panel.

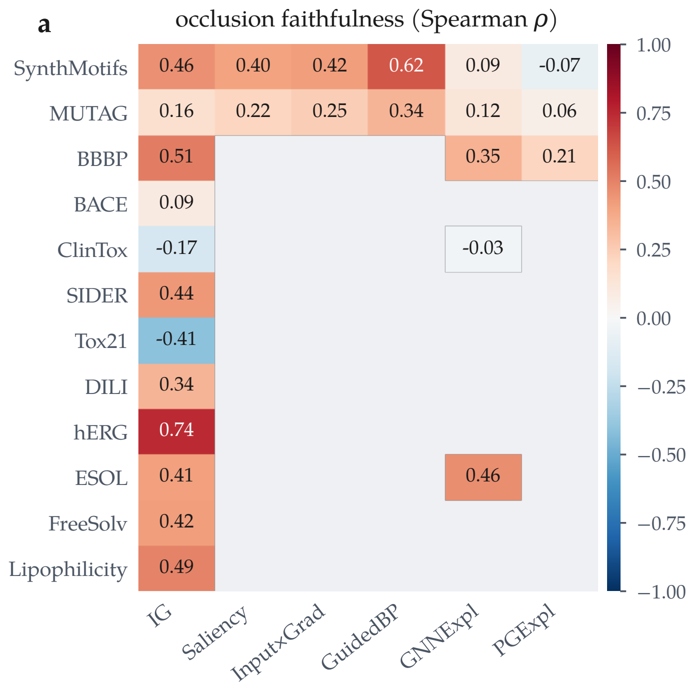
The arms that could not participate at all are the non-molecular ones. SynthMotifs, BA-2Motifs and ShapeGGen have exact node labels but are not molecules, so a Bemis–Murcko scaffold is undefined for them and their "scaffold" split comes back as an arbitrary deterministic partition (§). The shift contrast therefore rests on the 3 molecular arms and their 33 cells, two of which carry a ground truth that is exact by construction and one a chemically motivated proxy.

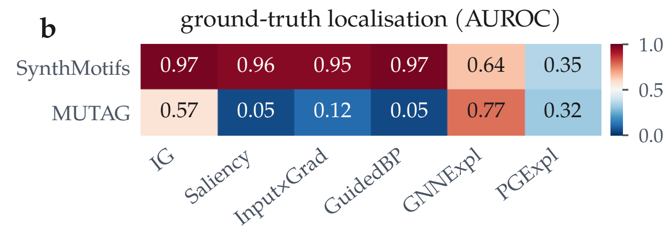
Two caveats. First, our own faithfulness metric fails alongside the field-standard ones on the MUTAG shift arm. We therefore do not claim that MolSanity's occlusion measure is a better selector. Under shift, *no* faithfulness metric in this experiment carries reliable information about which attributor is correct, which argues for measuring correctness rather than for building a better proxy. Second, the in-distribution verdict rests on a single split of a single dataset, and the agreement it reports is weaker evidence than the disagreement under shift: agreement is what the null predicts.

Reliability is a property of the cell

Figure 4 lays the two axes over the audited grid, and two further comparisons show that neither axis is a stable property of the attributor or of the architecture.

Backbones. Holding the attributor fixed at Integrated Gradients and varying the backbone (Figure 5) gives a spread of 0.197 AUROC in distribution (GINE 0.990 down to GAT 0.793) and 0.128 under shift (GINE 0.973 down to MPNN 0.845), comparable to the spread across attributors on the same dataset. The two orderings are negatively correlated (ρ = 0.100, 5 backbones): the architecture that localises the motif best in distribution is not the one that does so under shift. The attributor ordering is somewhat more robust across the same change of split (ρ = 0.964 over 7 attributors) but far from preserved.





Panel b stops after two rows.

Of the 12 datasets audited, 2 carry node-level labels against which an attribution can be scored. For the other 10, every real molecular property dataset in the sweep, no such labels are published, so only panel a can be computed at all. Faithfulness is auditable everywhere; correctness almost nowhere.

Figure 4: Reliability is a property of the (attributor, dataset) pair, not of the attributor. All committed GINE cells at the scaffold split. (**a**) Occlusion faithfulness over every audited dataset; blue = anti-faithful. (**b**) Ground-truth localisation, which exists for two datasets only. Panel b is short because correctness is unmeasurable elsewhere, not because it scores badly there: no molecular dataset in the sweep publishes per-atom labels. Blank cells were not audited; a thin outline marks a value carried from an earlier run.

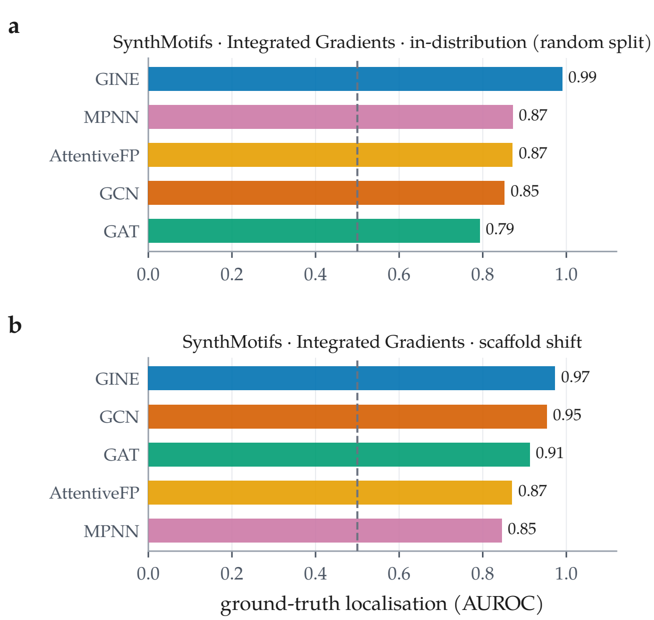


Figure 5: The backbone ordering does not survive the change of split. Integrated Gradients held fixed; ground-truth localisation across the 5 backbones on SynthMotifs, in distribution (**a**) and under scaffold shift (**b**). Dashed line is chance. Spearman correlation between the two orderings is 0.100 over 5 backbones.

The shift itself does not lower faithfulness. It would be tidy if scaffold shift simply made attributions less faithful. It does not. Across the 79 cells this run audited on both splits, occlusion faithfulness falls under the scaffold split in 35 of them, roughly half, with a median change of 0.048 (Figure 6b). We reported the opposite trend in an earlier, partial version of this matrix; with the grid complete it does not hold. What shift damages is the *relationship* between faithfulness and correctness, not the level of faithfulness, which is why a faithfulness metric gives no warning that anything has changed.

Does the model actually read the rationale?

The standing objection to every result above is Faber et al.'s [13]: scoring an attribution against a labelled substructure is misleading if the trained model does not use that substructure, in which case a low GT AUROC is a fact about the model rather than about the explanation. The objection is correct, and it is testable rather than merely arguable. For each molecule we occlude the ground-truth substructure and record whether the prediction collapses; if it does, the model demonstrably reads that substructure, and its attribution can be judged against it on the model's own behaviour.

The objection applies to more molecules than it does not. 7,667 of the audited molecules carry a rationale the model does not rely on, against 5,366 that it does, and the split is visible in the scores: mean GT AUROC is 0.811 where the model reads the rationale and 0.722 where it does not. A framework that ignored this distinction would be scoring the model under the guise of scoring the explanation, on roughly three molecules in five.

The number that settles it is what remains. Among the 5,366 molecules whose prediction demonstrably depends on the labelled substructure, 837 — a fraction of 0.156 — receive an attribution *anti-aligned* with it: the substructure the model provably uses is ranked among the atoms the attribution deems least relevant. On those molecules no appeal to an alternative rationale is available. The attribution misdescribes a model by that model's own behaviour, and faithfulness metrics computed on the same molecules do not flag it. We therefore treat the Faber objection as a partition to report rather than a defeater: it removes a majority of molecules from the correctness argument, and what survives it is still a large, unambiguous population of wrong explanations.

Where attributions fail: regimes and calibration

With per-molecule records available, two audit axes that previous drafts could only define are now measurable (Figure 7).

Regime stratification. Pooling the 26185 audited molecules by regime gives the audit's most directly actionable result. Ground-truth localisation is highest on confident-correct molecules (0.790, $n = 8736$), drops to 0.696 on borderline ones ($n = 4040$), and is lowest, at 0.674, on *confident-error* molecules ($n = 257$). It stays above chance throughout, so the attribution does not invert wholesale; it degrades exactly where a deployment can least afford it, in the regime where an explanation is most likely to be believed.

The two cheap proxies both point the wrong way there. Occlusion faithfulness on confident-error molecules is 0.330, markedly *better* than the 0.126 of confident-correct ones, and cross-checkpoint stability is 0.829 against 0.805 — also no worse. A practitioner screening explanations by faithfulness or stability would rank the least correct group at or above the most correct one. The confident-error estimate rests on 257 molecules and is reported with that n throughout; it is suggestive rather than precise.

Calibration linkage. The framework's hypothesis is that better-calibrated predictions carry more trustworthy attributions. Computed per cell it is supported on balance (median $\rho = 0.137$ across 146 classification cells, 69 significantly positive against 26 significantly negative), but the effect is cell-specific rather than universal. Pooling every molecule into a single correlation instead yields 0.046, roughly a quarter of the per-cell median and indistinguishable from no relationship at all. The pooled number is an artefact of mixing cells with different confidence distributions, and we report it only to warn against it: an evaluation that reports one aggregate correlation would conclude the linkage does not exist, when per cell it does.

Cross-task reliability

Tables 3, 7 and 8 give every audited combination on a real molecular dataset. Three cross-task patterns hold, and one apparent pattern turns out to be an artefact.

The regression faithfulness operator was mis-specified, and we withdraw its numbers. Across the 12 regression combinations, 0 have negative occlusion faithfulness (median 0.479), on cells whose predictive performance is well behaved (R^2 from 0.740 to 0.867 on ESOL). We do not interpret that pattern, and on inspection it is an artefact of our own operator rather than a property of the attributions. The operator compared two different quantities: the attribution was clipped at zero, keeping only atoms that push the prediction *up*, while the occlusion effect kept its sign. A motif that drives a solubility prediction strongly down therefore scored as unimportant on the attribution side and dominant on the occlusion side, which is sufficient on its own to produce a systematic negative correlation. The unbounded output-space Fidelity$\pm$ values, which reach 7.50 against the $[-1, 1]$ a probability-space fidelity occupies, are the visible symptom of the same mismatch. The corrected operator ranks by magnitude on both sides, reports the GraphFramEx characterisation score as undefined rather than clipping a standard-deviation-space shift into $[0, 1]$, and adds a bounded statistic giving the share of the total occlusion effect carried by the salient atoms. It is implemented and tested but postdates this sweep, so Table 3 is retained for completeness and the regression cells are excluded from every faithfulness claim in this paper. The classification path, on which every other result rests, is unchanged.

Coherence is high nearly everywhere, and says little. Motif top-1 share lies between 0.081 and 1.000 across the 124 molecular cells, with 82 of them inside a narrow 0.6–0.9 band, largely independent of whether the same cell is faithful or anti-faithful. Concentration on a single motif is easy to achieve and, on this evidence, close to uninformative about reliability. It belongs in the audit as a descriptive statistic, not as a quality score.

Stability is high but not uniform. Median cross-checkpoint stability over 152 cells is 0.847, with a minimum of -0.297, a cell whose early and final attributions are essentially uncorrelated. By attributor, GNNExplainer is the most stable (median 0.946) and Integrated

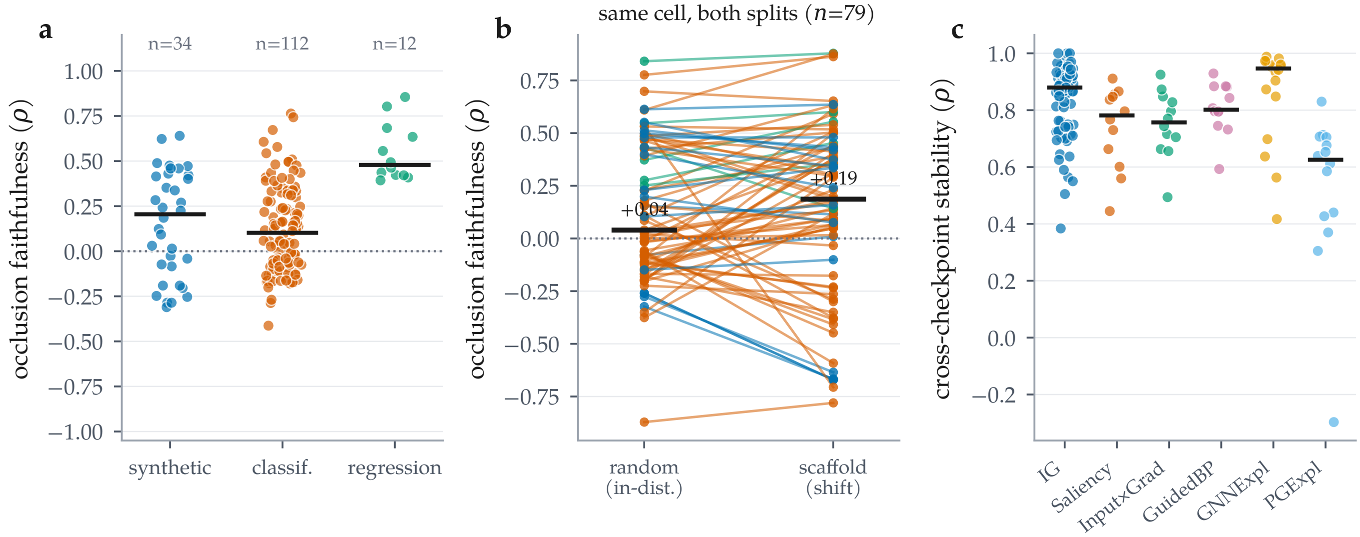


Figure 6: Distributions across all audited combinations. (**a**) Occlusion faithfulness by dataset regime; bars are medians. (**b**) The same cell on both splits, joined: faithfulness falls under the scaffold split in only 35 of 79 cells (median change 0.048), so shift does not systematically reduce faithfulness. (**c**) Cross-checkpoint stability by attributor; the mask-learning methods are not uniformly more stable than the gradient ones.

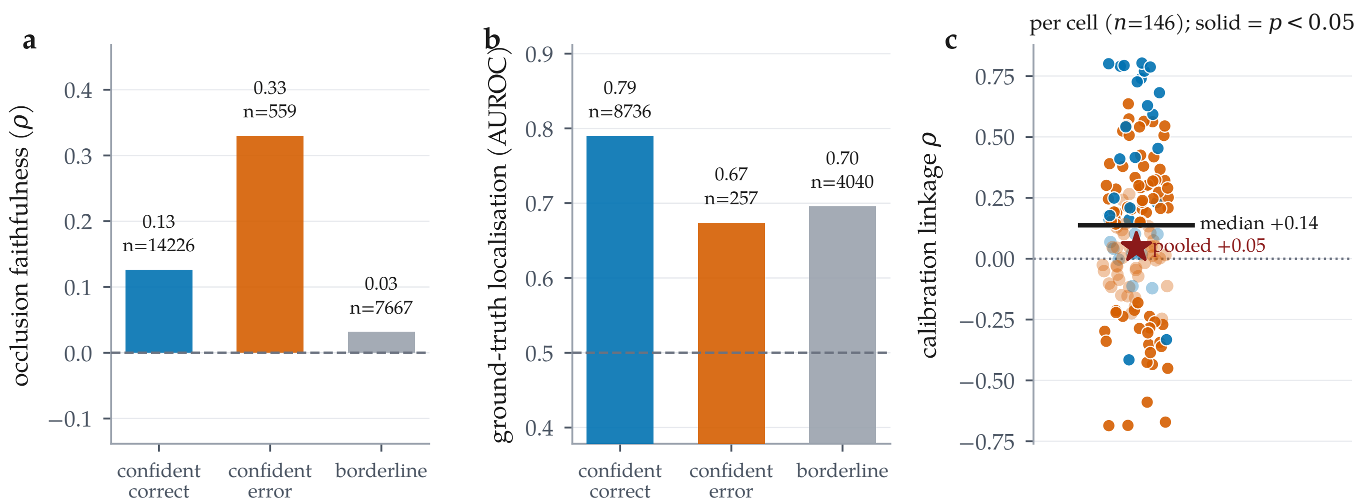


Figure 7: Regime stratification and calibration linkage, over the 26185 per-molecule audit records. (**a**) Occlusion faithfulness and (**b**) ground-truth localisation by confidence/correctness regime; n is printed per bar because ground truth exists for only a minority of molecules. Localisation on confident-*error* molecules degrades (0.790 to 0.674, $n = 257$) while occlusion faithfulness on the same molecules *improves* (0.126 to 0.330). (**c**) Calibration linkage computed per cell (solid = $p < 0.05$): the median cell is clearly positive, but pooling all molecules into one correlation attenuates it to near zero, a Simpson's paradox and a caution against reporting the pooled number.

Gradients sits at 0.879 over 80 cells; the per-attributor counts are very unequal in this run, so these medians are descriptive rather than a controlled comparison.

Table 4 gives the paired attributor contrasts behind Section A fidelity-only ranking picks the wrong attributor under shift, computed from the per-molecule records.

Negative and degenerate results

We record what did not work alongside what did.

Ground truth that was not being read. BA-2Motifs is a synthetic benchmark whose whole purpose is to carry exact node labels. It trains in this run and produces 2×100 audited molecules, but both its cells sit in the no-ground-truth block, because the loader found no node labels to read. The cause was ours rather than the dataset's: PyG's `BA2MotifDataset` exposes only node features, an edge index and a graph-level label, and our extractor looked for a per-node field that is never populated. The labels are recoverable from the released node ordering, in which the injected motif is appended after the Barabási–Albert base. That recovery is now implemented and verified structurally, by requiring the induced subgraph on those nodes to be a house or a five-cycle before the mask is used, and cross-checked against a generator that produces the same construction with independent labels. It is refused rather than guessed when the check fails, so a reordered loader degrades to the behaviour reported here. The numbers in this paper predate that fix: BA-2Motifs is counted as audited and not as a ground-truth cell throughout, and the third exact-ground-truth arm it will provide is not claimed here.

The best model has the worst attributions. The highest-AUC molecular classification cell in the run is MolMotif · AttentiveFP under the scaffold split, at accuracy 0.985, ROC-AUC 1.000 and ECE 0.011, so accurate and well calibrated. Its attributions are the most *anti*-faithful of any molecular cell in the matrix ($\rho = 0.672$): Integrated Gradients ranks the motifs of these molecules in close to the reverse of their causal order. Predictive quality, calibration and explanation reliability are three different axes.

A weak model can be faithfully explained. The converse also appears. The most faithful molecular cell is MUTAG · GCN ($\rho = 0.764$) on a model with accuracy 0.799 and ROC-AUC 0.894, barely better than chance. Faithfulness certifies that the explanation describes the model; it says nothing about whether the model is worth describing. Any reliability report that omits the predictive metrics can be satisfied by this cell.

Accuracy is the wrong headline on imbalanced sets. 30 committed rows report test accuracy $\geq$ 0.95, with ROC-AUC as low as 0.981. On Tox21, accuracy runs 0.940–0.942 against AUC 0.745–0.820. We report AUC alongside accuracy throughout for this reason, and the lowest-AUC cell in the run (ShapeGGen · GINE, AUC 0.552 at accuracy 0.580) is exactly the case where accuracy alone would mislead.

Blocked, and re-examined. The results reported here come from a run in which SubgraphX (via DIG) and ShapeGGen (via GraphXAI) were both recorded as uninstallable, on the grounds that they need the compiled `torch_sparse/torch_scatter` extensions for which no wheel exists at the pinned torch version. Revisiting that diagnosis, it was wrong in both cases. Those extensions build from source, DIG then installs, and SubgraphX is now wrapped and covered by tests in which it recovers a planted five-node motif exactly; it is enabled in the configuration and will populate the matrix on the next sweep. GraphXAI installs from a source checkout rather than a release, and ShapeGGen builds. It remains unintegrated for a different and more substantial reason: ShapeGGen is a node-classification benchmark on a single large graph, whereas every axis of this audit is defined per molecule at the graph level, so adopting it means extending the audit to node-level tasks rather than resolving a dependency. Both are recorded in the ledger, occupy no row in any table, and enter no aggregate. We report the earlier diagnosis alongside the correction because a coverage gap attributed to the environment turned out to be a gap in how hard we looked.

What to do when there is no ground truth

Every result above depends on rationale labels, which is exactly what a working medicinal chemist does not have. If the audit only says "your attributions may be wrong and you cannot check", it has diagnosed a problem and left the reader where it found them. The weaker but usable question is not *which attributor should I trust* but *on which molecules should I decline to trust this one* — a selective-prediction framing, transplanted from prediction to explanation. We rank 5 candidate signals, all computable at inference time without any labels, by the gain in mean ground-truth localisation between full coverage and half coverage (Figure 8).

The best is predicted confidence (+0.058 over 13,033 molecules with ground truth). Ranking by it and keeping the most confident 10% cuts the share of retained molecules whose attribution is *below chance* from 0.203 to 0.098, with mean localisation 0.829 over the 1,303 retained ($n = 1,303$). We state the rule against the below-chance share rather than the mean deliberately: pooled mean localisation is already tolerable at full coverage, so a rule stated against it would look successful while leaving the failures untouched. The tail is what a deployment gets hurt by, and the tail is what this moves.

The ranking's negative result is as useful as its positive one. 1 of the 5 signals has *negative* lift: motif top-1 share (−0.090). Abstaining on molecules whose attribution mass is spread across many motifs — keeping those that concentrate on a single chemically coherent one — makes the retained set *worse*. That is the intuition a chemist is most likely to apply by eye, and on this data it is anti-predictive. An audit that reported only the working signal would have left that trap in place.

What this rule does not license. These curves are fitted where ground truth exists. Applying them to a dataset where it does not assumes the signal–reliability relationship transfers, and the central finding of this paper is that one such relationship — faithfulness to correctness — does not survive a change of split. We therefore offer

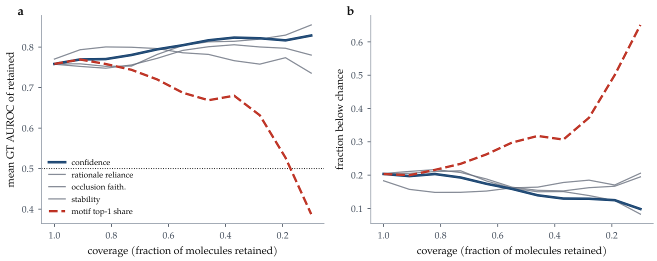


Figure 8: Coverage–reliability curves for abstention signals, over the $13,033$ **audited molecules that carry ground truth.** Every signal is computable at inference time without rationale labels. Moving left to right is abstaining on more molecules. (**a**) Mean ground-truth localisation of the retained set; the dotted line is chance. (**b**) The fraction of the retained set whose attribution is *below* chance — the quantity a deployment is hurt by, and the one the pooled mean in (a) hides. The best signal (predicted confidence) is highlighted; motif top-1 share is dashed because its lift is negative, so abstaining by it makes the retained set worse.

the rule as a calibrated starting point that a group can re-fit on its own labelled subset, not as a constant of nature. Stating that constraint is not hedging: it is the same standard we hold the faithfulness metrics to.

Discussion

What the audit buys. The single most useful thing MolSanity produces is not a score but a *negative certificate*. For 44 of the 158 committed rows, covering every real molecular dataset in the sweep, attribution correctness cannot be measured at all, because no per-atom explanation labels exist. Those cells can be audited for faithfulness, coherence, stability and calibration, and this paper's central result is that under shift none of those is a reliable stand-in for correctness. Making that gap explicit is more useful than filling it with a metric that looks like correctness and is not.

How attributions should be reported. Four practices follow directly from the matrix. (i) Report the regime: an attributor selected in distribution carries no warrant out of it, and on the one cell family where we can check, the selection is significantly wrong under shift. (ii) Report the backbone: the backbone ordering does not survive the change of split ($\rho = 0.100$), so "IG on a GNN" is under-specified. (iii) Report the predictive metrics next to the explanation metrics, because the accurate, well-calibrated MolMotif · AttentiveFP cell has the most anti-faithful attributions in the matrix. (iv) Do not pool: the calibration-linkage correlation changes sign between the per-cell and the pooled analysis.

Why reliability is regime-, backbone- and dataset-dependent. A faithfulness metric asks about the model; a ground-truth metric asks about the chemistry; they agree only when the model has learned the chemistry. That coincidence is engineered into synthetic benchmarks and is not guaranteed on molecular data, particularly out of distribution, which is the situation Faber et al. [13] identify as making ground-truth comparison hazardous in general. Attribution evaluation inherits the model's inductive limitations, so the correct object of study is the cell, (dataset, backbone, attributor, split), and not the attributor alone.

Limitations and threats to validity

Ground truth is scarce, and one arm of it is a proxy. The two cell families that carry the argument have $n = 53$ audited molecules per arm. MUTAG, the molecular one, uses a chemically motivated nitro/aromatic-nitro SMARTS proxy rather than annotator labels: it encodes an assumption about which substructure ought to matter, and the trained model may use a different, equally valid rationale [13]. We answer that objection empirically rather than by assertion (§), but it remains the case that the shift contrast rests on this one dataset, at $n = 53$ audited molecules per arm, with a proxy ground truth.

The arms that might have corroborated it cannot. SynthMotifs, BA-2Motifs and ShapeGGen have exact node labels but are not molecules, so a Bemis–Murcko scaffold is undefined for them: our splitter reports every graph in its own scaffold bucket, which makes their "scaffold" split an arbitrary deterministic partition rather than a chemical shift, and we exclude them from the contrast rather than let a degenerate partition stand in for one. MolMotif and MolMotifHard are molecular *and* exactly labelled, by construction, since the label is presence of a substructure. Their limitation is not that they are saturated outright but that their attributor rankings are *bimodal*: four of seven sit above GT AUROC 0.9 within a few hundredths of one another, so a seven-point rank correlation on either arm is carried by a handful of comparisons among effectively tied methods. We built MolMotifHard to widen that range and it does, but not enough to make either arm decisive alone. No public dataset is simultaneously molecular, exactly labelled and *graded* in difficulty; constructing one is the clearest next step for this line of work, and until it exists the pooled estimate across arms is a better instrument than any single one of them.

A dataset that trains but was not scored. BA-2Motifs loads and trains in this run, but its cells contribute faithfulness only and appear in the no-ground-truth block despite being a synthetic benchmark designed to have exact labels, because our extractor did not read the ordering its node labels are encoded in. The recovery is implemented and tested but postdates this sweep, so the third exact-ground-truth arm is available to the next run and is not part of any number reported here.

Single split, single seed. Each cell is one deterministic split at one seed. Cross-checkpoint stability partially probes run-to-run variability, but no confidence interval in this paper covers split or seed variance, and the per-cell molecule counts (20 on the synthetic arms) make individual contrasts underpowered.

Occlusion as a counterfactual. Zeroing node features takes the graph off the data manifold; a motif whose removal barely moves the output may be redundant rather than unimportant. This bears most heavily on the regression cells, where the output-space Fidelity$\pm$ values leave the range a probability-space fidelity occupies and we decline to interpret the sign.

Proxy ground truth. MUTAG's mask is a nitro/aromatic-nitro SMARTS proxy, not annotator labels. It encodes an assumption about which substructure ought to matter, and the trained model may use a different, equally valid rationale [13].

Coverage of attributors. SubgraphX contributes no row to this run and PGExplainer failed in it, so every conclusion about the perturbation family rests on GNExplainer alone in the current-run cells. SubgraphX is wrapped and tested but its cells postdate this sweep, so the gap stands for the numbers reported here.

Multiplicity. The selection tests carry Benjamini–Hochberg adjusted q values over the family of 18 tests, and the attributor contrasts carry them over the family within each cell block, so the headline mismatches are false-discovery-rate controlled rather than descriptive. What the adjustment cannot repair is power: at $n = 53$ molecules per arm a contrast that fails to reach significance is weak evidence of no difference, not evidence of none.

Conclusion

Molecular GNN attributions are audited today for whether they describe the model. Across the committed audit matrix we find that this is not the same question as whether they describe the chemistry, and that the two diverge in the regime molecular property prediction actually operates in. On the one cell family where exact ground truth, several attributors and both splits coexist, a faithfulness-only ranking is harmless in distribution and selects a significantly worse attributor under scaffold shift. MolSanity's own faithfulness metric is no exception. Reliability does not transfer across backbones or across splits, and for the 44 molecular rows in the matrix, correctness cannot be measured at all with the labels the field currently has. The audit is what makes those failures visible; the pipeline, the figures and every number here regenerate from the committed artifacts as the grid fills in.

Data and code availability

All code, configurations, trained checkpoints, per-combination results, per-molecule audit records and the run manifest (seed, library versions, hardware, and the per-combination outcome ledger) are available at `https://github.com/Kar488/molsanity`, released under the MIT licence. A versioned archive of the release, including the trained weights, is deposited with a DOI [*DOI to be inserted on deposition*]; the repository carries `CITATION.cff` and `.zenodo.json` so the archive is generated from a tagged release.

Every figure and table in this paper is generated from those artifacts by `paper/figs/make_figures.py` and `paper/figs/make_tables.py`, which read the released results directly; no value in this manuscript is transcribed by hand. Re-running them after a new sweep updates every number, figure and table in place. Dataset licences and provenance are recorded in the repository's dataset manifest; the credential-gated resources referenced in Section Experimental setup are out of scope and were never accessed.

Table 4: Paired attributor comparisons on shared molecules, computed from the committed per-molecule records (Wilcoxon signed-rank on per-molecule occlusion faithfulness, $\Delta = A - B$). Same cell family as Table 5. p is the raw two-sided Wilcoxon value and q its Benjamini–Hochberg adjustment, controlling the false discovery rate over the contrasts within each cell block, which is the family a reader compares across.

A vs. B	n	**median** Δ	p	q
MUTAG, GINE, random split				
IG vs. Saliency	58	0.000	0.885	0.885
IG vs. Input×Grad	58	0.000	0.862	0.885
IG vs. GuidedBP	58	0.000	0.046	0.108
IG vs. GNNExpl.	58	0.000	0.704	0.861
IG vs. PGExpl.	57	0.149	0.006	0.019
IG vs. SubgraphX	57	0.000	0.692	0.861
Saliency vs. Input×Grad	58	0.000	0.779	0.861
Saliency vs. GuidedBP	58	0.000	0.005	0.018
Saliency vs. GNNExpl.	58	0.000	0.597	0.861
Saliency vs. PGExpl.	57	0.193	<0.001	0.001
Saliency vs. SubgraphX	57	0.000	0.689	0.861
Input×Grad vs. GuidedBP	58	0.000	0.005	0.018
Input×Grad vs. GNNExpl.	58	0.000	0.608	0.861
Input×Grad vs. PGExpl.	57	0.213	<0.001	0.001
Input×Grad vs. SubgraphX	57	−0.015	0.711	0.861
GuidedBP vs. GNNExpl.	58	0.046	0.148	0.294
GuidedBP vs. PGExpl.	57	0.316	<0.001	<0.001
GuidedBP vs. SubgraphX	57	0.028	0.154	0.294
GNNExpl. vs. PGExpl.	57	0.070	0.007	0.019
GNNExpl. vs. SubgraphX	57	0.000	0.774	0.861
PGExpl. vs. SubgraphX	56	−0.167	<0.001	0.001
MUTAG, GINE, scaffold split				
IG vs. Saliency	53	0.000	0.098	0.171
IG vs. Input×Grad	53	−0.200	0.002	0.005
IG vs. GuidedBP	53	−0.343	<0.001	<0.001
IG vs. GNNExpl.	53	0.000	0.270	0.378
IG vs. PGExpl.	52	0.000	0.743	0.743
IG vs. SubgraphX	53	0.000	0.398	0.522
Saliency vs. Input×Grad	53	0.000	0.031	0.066
Saliency vs. GuidedBP	53	−0.107	<0.001	<0.001
Saliency vs. GNNExpl.	53	0.000	0.672	0.743
Saliency vs. PGExpl.	52	0.057	0.066	0.125
Saliency vs. SubgraphX	53	0.033	0.626	0.730
Input×Grad vs. GuidedBP	53	0.000	<0.001	0.001
Input×Grad vs. GNNExpl.	53	0.000	0.019	0.044
Input×Grad vs. PGExpl.	52	0.200	<0.001	0.002
Input×Grad vs. SubgraphX	53	0.023	0.130	0.195
GuidedBP vs. GNNExpl.	53	0.200	<0.001	<0.001
GuidedBP vs. PGExpl.	52	0.265	<0.001	<0.001
GuidedBP vs. SubgraphX	53	0.200	<0.001	<0.001
GNNExpl. vs. PGExpl.	52	0.022	0.118	0.191
GNNExpl. vs. SubgraphX	53	0.000	0.739	0.743
PGExpl. vs. SubgraphX	52	0.000	0.447	0.552

Table 5: Does a faithfulness-only ranking pick the ground-truth-best attributor? Two cell families, each audited on both splits with the same backbone and the same six attributors, so within a family only the split changes and the contrast isolates distribution shift rather than confounding it with a change of dataset. Each row ranks the attributors by one faithfulness/fidelity metric and asks whether its top choice is the one the ground truth ranks best. p is a paired Wilcoxon test on per-molecule GT AUROC between the selected and the ground-truth-best attributor, over the molecules both audited, and q its Benjamini–Hochberg adjustment over all 18 selection tests in this table.

cell	**regime**	**ranking metric**	**its top pick**	**pick GT**	**GT-best**	**GT-best**	**mismatch**	**Wilcoxon** p	q	ρ**(faith,GT)**
MUTAG·GINE, n=58	in-distribution	occlusion ρ	GuidedBP	0.037	GNNExpl.	0.858	**yes**	<0.001	<0.001	0.143
		Fidelity+	IG	0.496	GNNExpl.	0.858	**yes**	<0.001	<0.001	−0.357
		characterisation	SubgraphX	0.348	GNNExpl.	0.858	**yes**	<0.001	<0.001	0.821
MUTAG·GINE, n=53	scaffold shift	occlusion ρ	GuidedBP	0.013	GNNExpl.	0.826	**yes**	<0.001	<0.001	−0.643
		Fidelity+	SubgraphX	0.330	GNNExpl.	0.826	**yes**	<0.001	<0.001	−0.107
		characterisation	SubgraphX	0.330	GNNExpl.	0.826	**yes**	<0.001	<0.001	0.036
MolMotifHard·GINE, n=200	in-distribution	occlusion ρ	SubgraphX	0.421	Saliency	0.994	**yes**	<0.001	<0.001	−0.286
		Fidelity+	SubgraphX	0.421	Saliency	0.994	**yes**	<0.001	<0.001	0.107
		characterisation	SubgraphX	0.421	Saliency	0.994	**yes**	<0.001	<0.001	−0.357
MolMotifHard·GINE, n=200	scaffold shift	occlusion ρ	PGExpl.	0.565	IG	1.000	**yes**	<0.001	<0.001	−0.893
		Fidelity+	IG	1.000	IG	1.000	no	—	—	0.821
		characterisation	IG	1.000	IG	1.000	no	—	—	0.857
MolMotif·GINE, n=200	in-distribution	occlusion ρ	GNNExpl.	0.630	Input×Grad	0.999	**yes**	<0.001	<0.001	−0.571
		Fidelity+	GuidedBP	0.998	Input×Grad	0.999	**yes**	0.091	0.091	0.750
		characterisation	SubgraphX	0.556	Input×Grad	0.999	**yes**	<0.001	<0.001	−0.107
MolMotif·GINE, n=200	scaffold shift	occlusion ρ	IG	0.939	Input×Grad	1.000	**yes**	<0.001	<0.001	−0.286
		Fidelity+	IG	0.939	Input×Grad	1.000	**yes**	<0.001	<0.001	−0.321
		characterisation	IG	0.939	Input×Grad	1.000	**yes**	<0.001	<0.001	−0.286

Table 6: Every committed cell in which attribution *correctness* is measurable. Ground truth is exact for the synthetic sets and a chemically motivated nitro-motif proxy for MUTAG. GT AUROC = 0.5 is chance; below 0.5 the attribution is anti-aligned with the true motif. Occlusion ρ is the attribution–occlusion rank agreement (higher = more faithful to the model). Rows marked [c] are *carried*: they survive in the results matrix from an earlier reduced-budget CPU run because the corresponding cell failed in the latest run (Table 2); every other row was produced by the run in Table 2. Bold marks the best GT AUROC and the best occlusion ρ within each dataset×split block (emphasis only; values are unaltered).

dataset	backbone	attributor	split	n	acc	AUC	GT AUROC	GT AUPRC	occ. ρ	Fid+	Fid−	stab.
BA-2Motifs	GINE	GNNExpl.	random	200	0.767	1.000	0.511	0.308	−0.027	−0.001	0.019	0.951
BA-2Motifs	GINE	GuidedBP	random	200	0.767	1.000	0.945	0.901	−0.248	0.039	−0.006	0.745
BA-2Motifs	GINE	Input×Grad	random	200	0.767	1.000	0.975	0.908	−0.191	0.030	0.001	0.796
BA-2Motifs	GINE	IG	random	200	0.767	1.000	**0.997**	0.988	−0.204	0.030	0.001	0.645
BA-2Motifs	GINE	PGExpl.	random	200	0.767	1.000	0.398	0.304	**0.015**	0.006	0.027	0.305
BA-2Motifs	GINE	Saliency	random	200	0.767	1.000	0.975	0.908	−0.191	0.030	0.001	0.796
BA-2Motifs	GINE	GNNExpl.	scaffold	200	0.603	0.978	0.563	0.335	−0.041	−0.002	−0.007	0.969
BA-2Motifs	GINE	GuidedBP	scaffold	200	0.603	0.978	0.843	0.618	−0.253	0.000	−0.004	0.807
BA-2Motifs	GINE	Input×Grad	scaffold	200	0.603	0.978	**0.937**	0.789	−0.286	−0.001	−0.005	0.663
BA-2Motifs	GINE	IG	scaffold	200	0.603	0.978	0.856	0.543	−0.310	−0.003	−0.004	0.715
BA-2Motifs	GINE	PGExpl.	scaffold	200	0.603	0.978	0.886	0.768	**0.030**	−0.002	−0.005	−0.297
BA-2Motifs	GINE	Saliency	scaffold	200	0.603	0.978	0.937	0.789	−0.286	−0.001	−0.005	0.663
MUTAG	AttentiveFP	IG	random	58	0.735	0.915	0.040	0.171	0.156	0.065	0.124	0.822
MUTAG	GAT	IG	random	58	0.621	0.672	0.400	0.379	0.151	0.209	0.296	0.951
MUTAG	GCN	IG	random	58	0.638	0.700	0.367	0.368	**0.496**	0.237	0.302	0.911
MUTAG	GINE	GNNExpl.	random	58	0.747	0.912	**0.678**	0.590	−0.121	0.189	0.205	0.848
MUTAG	GINE	GuidedBP	random	58	0.747	0.912	0.097	0.197	−0.131	0.330	0.351	0.882
MUTAG	GINE	Input×Grad	random	58	0.747	0.912	0.035	0.169	−0.172	0.295	0.354	0.770
MUTAG	GINE	IG	random	58	0.747	0.912	0.387	0.356	−0.177	0.330	0.313	0.724
MUTAG	GINE	PGExpl.	random	58	0.747	0.912	0.663	0.600	−0.159	0.246	0.292	0.713
MUTAG	GINE	Saliency	random	58	0.747	0.912	0.012	0.165	−0.151	0.294	0.356	0.767
MUTAG	GINE	SubgraphX	random	58	0.747	0.912	0.393	0.308	−0.035	0.276	0.029	—
MUTAG	MPNN	IG	random	58	0.787	0.722	0.103	0.181	0.358	0.506	0.516	0.890
MUTAG	AttentiveFP	IG	scaffold	53	0.774	0.886	0.029	0.134	0.325	0.008	0.017	0.907
MUTAG	GAT	IG	scaffold	53	0.824	0.913	0.533	0.506	0.607	0.054	0.171	1.000
MUTAG	GCN	IG	scaffold	53	0.799	0.894	0.083	0.164	**0.764**	0.159	0.340	0.941
MUTAG	GINE	GNNExpl.	scaffold	53	0.736	0.808	**0.774**	0.695	0.124	0.116	0.145	0.904
MUTAG	GINE	GuidedBP	scaffold	53	0.736	0.808	0.047	0.143	0.337	0.266	0.129	0.929
MUTAG	GINE	Input×Grad	scaffold	53	0.736	0.808	0.116	0.154	0.247	0.229	0.195	0.828
MUTAG	GINE	IG	scaffold	53	0.736	0.808	0.568	0.449	0.164	0.160	0.227	0.810
MUTAG	GINE	PGExpl.	scaffold	53	0.736	0.808	0.319	0.204	0.060	0.038	0.203	0.612
MUTAG	GINE	Saliency	scaffold	53	0.736	0.808	0.051	0.143	0.218	0.226	0.174	0.856
MUTAG	GINE	SubgraphX	scaffold	53	0.736	0.808	0.392	0.279	0.199	0.240	0.023	—
MUTAG	MPNN	IG	scaffold	53	0.837	0.913	0.163	0.185	0.206	0.166	0.253	0.829
MolMotif	AttentiveFP	IG	random	200	0.955	0.996	0.967	0.923	**0.496**	0.469	0.025	0.912
MolMotif	GAT	IG	random	200	1.000	1.000	0.842	0.776	0.338	0.440	0.225	0.901
MolMotif	GCN	IG	random	200	0.872	0.996	0.943	0.895	0.167	0.324	0.497	1.000
MolMotif	GINE	GNNExpl.	random	200	0.928	1.000	0.641	0.437	−0.017	0.184	0.296	0.973
MolMotif	GINE	GuidedBP	random	200	0.928	1.000	0.971	0.928	−0.113	0.205	0.293	0.884
MolMotif	GINE	Input×Grad	random	200	0.928	1.000	**0.995**	0.989	−0.179	0.146	0.300	0.848
MolMotif	GINE	IG	random	200	0.928	1.000	0.951	0.875	−0.004	0.341	0.218	0.970
MolMotif	GINE	PGExpl.	random	200	0.928	1.000	0.502	0.335	−0.098	0.132	0.302	0.639
MolMotif	GINE	Saliency	random	200	0.928	1.000	0.994	0.987	−0.164	0.161	0.300	0.837
MolMotif	GINE	SubgraphX	random	200	0.928	1.000	0.555	0.327	−0.036	0.215	0.215	—
MolMotif	MPNN	IG	random	200	1.000	1.000	0.873	0.729	−0.124	0.377	0.347	0.946
MolMotif	AttentiveFP	IG	scaffold	200	0.985	1.000	0.923	0.870	**0.672**	0.447	0.024	0.929
MolMotif	GAT	IG	scaffold	200	0.995	1.000	0.798	0.723	0.332	0.449	0.454	0.945
MolMotif	GCN	IG	scaffold	200	0.970	1.000	0.988	0.975	0.391	0.399	0.177	0.933
MolMotif	GINE	GNNExpl.	scaffold	200	0.973	1.000	0.616	0.425	0.054	0.301	0.344	0.982
MolMotif	GINE	GuidedBP	scaffold	200	0.973	1.000	0.989	0.977	0.033	0.284	0.346	0.843
MolMotif	GINE	Input×Grad	scaffold	200	0.973	1.000	**0.999**	0.997	−0.086	0.156	0.360	0.716
MolMotif	GINE	IG	scaffold	200	0.973	1.000	0.897	0.749	0.058	0.376	0.313	0.880
MolMotif	GINE	PGExpl.	scaffold	200	0.973	1.000	0.513	0.361	−0.111	0.129	0.299	0.707
MolMotif	GINE	Saliency	scaffold	200	0.973	1.000	0.997	0.991	−0.090	0.139	0.367	0.866
MolMotif	GINE	SubgraphX	scaffold	200	0.973	1.000	0.469	0.265	−0.007	0.362	0.129	—
MolMotif	MPNN	IG	scaffold	200	0.998	1.000	0.871	0.727	0.055	0.399	0.348	0.922
MolMotifHard	AttentiveFP	IG	random	200	0.972	0.981	0.878	0.823	**0.476**	0.411	0.087	0.932
MolMotifHard	GAT	IG	random	200	0.962	0.992	0.793	0.770	0.041	0.466	0.404	0.966
MolMotifHard	GCN	IG	random	200	0.885	0.988	0.901	0.855	0.040	0.359	0.355	0.700
MolMotifHard	GINE	GNNExpl.	random	200	0.935	0.998	0.413	0.385	−0.075	0.141	0.251	0.942
MolMotifHard	GINE	GuidedBP	random	200	0.935	0.998	0.919	0.890	−0.050	0.204	0.243	0.884
MolMotifHard	GINE	Input×Grad	random	200	0.935	0.998	0.996	0.994	−0.068	0.228	0.221	0.925
MolMotifHard	GINE	IG	random	200	0.935	0.998	0.985	0.973	−0.074	0.218	0.231	0.914
MolMotifHard	GINE	PGExpl.	random	200	0.935	0.998	0.512	0.419	−0.051	0.162	0.208	0.830
MolMotifHard	GINE	Saliency	random	200	0.935	0.998	**0.998**	0.996	−0.064	0.217	0.217	0.911
MolMotifHard	GINE	SubgraphX	random	200	0.935	0.998	0.408	0.320	−0.040	0.201	0.150	—
MolMotifHard	MPNN	IG	random	200	0.908	0.996	0.953	0.920	0.310	0.289	0.222	0.933
MolMotifHard	AttentiveFP	IG	scaffold	200	0.975	0.995	0.900	0.825	0.316	0.388	0.075	0.893
MolMotifHard	GAT	IG	scaffold	200	0.958	0.997	0.807	0.737	−0.045	0.382	0.356	0.921
MolMotifHard	GCN	IG	scaffold	200	0.845	0.995	0.866	0.761	0.026	0.261	0.401	0.637
MolMotifHard	GINE	GNNExpl.	scaffold	200	0.892	0.997	0.420	0.349	−0.167	0.180	0.225	0.960
MolMotifHard	GINE	GuidedBP	scaffold	200	0.892	0.997	0.875	0.818	−0.136	0.285	0.186	0.794
MolMotifHard	GINE	Input×Grad	scaffold	200	0.892	0.997	0.995	0.990	−0.173	0.273	0.199	0.873
MolMotifHard	GINE	IG	scaffold	200	0.892	0.997	0.994	0.986	−0.201	0.308	0.188	0.909
MolMotifHard	GINE	PGExpl.	scaffold	200	0.892	0.997	0.376	0.300	−0.087	0.087	0.235	0.705
MolMotifHard	GINE	Saliency	scaffold	200	0.892	0.997	**0.999**	0.995	−0.165	0.278	0.211	0.847
MolMotifHard	GINE	SubgraphX	scaffold	200	0.892	0.997	0.472	0.310	−0.130	0.238	0.175	—
MolMotifHard	MPNN	IG	scaffold	200	0.987	0.998	0.965	0.924	**0.578**	0.480	0.093	0.886
ShapeGGen	GINE	GNNExpl.	random	50	0.780	0.578	0.549	0.591	0.389	0.239	0.155	0.563
ShapeGGen	GINE	GuidedBP	random	50	0.780	0.578	**0.773**	0.811	0.481	0.239	0.161	0.796
ShapeGGen	GINE	Input×Grad	random	50	0.780	0.578	0.725	0.764	0.435	0.250	0.171	0.744
ShapeGGen	GINE	IG	random	50	0.780	0.578	0.733	0.771	0.447	0.266	0.154	0.763
ShapeGGen	GINE	PGExpl.	random	50	0.780	0.578	0.499	0.523	−0.163	0.127	0.299	0.676
ShapeGGen	GINE	Saliency	random	50	0.780	0.578	0.769	0.808	0.400	0.228	0.191	0.730
ShapeGGen	GINE	SubgraphX	random	50	0.780	0.578	0.644	0.592	**0.543**	0.403	−0.010	0.233
ShapeGGen	GINE	GNNExpl.	scaffold	50	0.580	0.552	0.574	0.628	0.143	0.028	0.020	0.417
ShapeGGen	GINE	GuidedBP	scaffold	50	0.580	0.552	**0.803**	0.826	0.227	0.027	0.015	0.593
ShapeGGen	GINE	Input×Grad	scaffold	50	0.580	0.552	0.708	0.735	0.143	0.021	0.022	0.494
ShapeGGen	GINE	IG	scaffold	50	0.580	0.552	0.727	0.749	0.111	0.025	0.019	0.505
ShapeGGen	GINE	PGExpl.	scaffold	50	0.580	0.552	0.567	0.611	−0.062	0.021	0.034	0.653
ShapeGGen	GINE	Saliency	scaffold	50	0.580	0.552	0.776	0.801	0.164	0.020	0.025	0.445
ShapeGGen	GINE	SubgraphX	scaffold	50	0.580	0.552	0.598	0.580	**0.407**	0.059	−0.006	0.217
SynthMotifs	AttentiveFP	IG	random	200	0.980	0.999	0.871	0.721	0.185	0.441	0.230	0.726
SynthMotifs	GAT	IG	random	200	0.990	1.000	0.793	0.660	0.242	0.284	0.209	0.384
SynthMotifs	GCN	IG	random	200	0.888	1.000	0.852	0.767	0.415	0.323	0.125	0.694
SynthMotifs	GINE	GNNExpl.	random	200	0.972	1.000	0.632	0.388	0.123	0.232	0.251	0.698
SynthMotifs	GINE	GuidedBP	random	200	0.972	1.000	**0.995**	0.983	**0.640**	0.272	0.061	0.795
SynthMotifs	GINE	Input×Grad	random	200	0.972	1.000	0.970	0.899	0.475	0.273	0.111	0.656
SynthMotifs	GINE	IG	random	200	0.972	1.000	0.990	0.960	0.472	0.265	0.095	0.849
SynthMotifs	GINE	PGExpl.	random	200	0.972	1.000	0.415	0.209	−0.084	0.091	0.226	0.427
SynthMotifs	GINE	Saliency	random	200	0.972	1.000	0.979	0.927	0.489	0.274	0.105	0.560
SynthMotifs	GINE	SubgraphX	random	200	0.972	1.000	0.770	0.533	0.459	0.245	0.056	0.744
SynthMotifs	MPNN	IG	random	200	0.993	1.000	0.872	0.694	0.337	0.423	0.417	0.590
SynthMotifs	AttentiveFP	IG	scaffold	200	0.963	1.000	0.870	0.688	0.270	0.375	0.204	0.741
SynthMotifs	GAT	IG	scaffold	200	0.872	1.000	0.912	0.820	0.430	0.496	0.269	1.000
SynthMotifs	GCN	IG	scaffold	200	0.860	0.996	0.953	0.826	0.352	0.246	0.144	0.625
SynthMotifs	GINE	GNNExpl.	scaffold	200	0.947	1.000	0.642	0.397	0.092	0.327	0.341	0.637
SynthMotifs	GINE	GuidedBP	scaffold	200	0.947	1.000	**0.974**	0.928	**0.623**	0.424	0.045	0.732
SynthMotifs	GINE	Input×Grad	scaffold	200	0.947	1.000	0.955	0.865	0.421	0.403	0.133	0.705
SynthMotifs	GINE	IG	scaffold	200	0.947	1.000	0.973	0.901	0.459	0.412	0.065	0.741
SynthMotifs	GINE	PGExpl.	scaffold	200	0.947	1.000	0.354	0.181	−0.073	0.105	0.339	0.585
SynthMotifs	GINE	Saliency	scaffold	200	0.947	1.000	0.957	0.886	0.401	0.394	0.151	0.601
SynthMotifs	GINE	SubgraphX	scaffold	200	0.947	1.000	0.751	0.500	0.283	0.377	0.045	0.764
SynthMotifs	MPNN	IG	scaffold	200	0.838	0.999	0.845	0.651	0.225	0.317	0.288	0.550

Table 7: Molecular classification cells (no node-level ground truth available), 1 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets, since no per-atom labels are published, so only model-side reliability can be measured, which is the gap the Tier-1 cells are needed to close. *charact.* is the GraphFramEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules; both are blank for carried rows (^{c}), whose per-molecule records this run did not regenerate. Bold marks the highest occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid–	stab.	charact.	unfaith.
BACE	GCN	IG	random	200	0.720	0.819	0.052	0.771	**0.235**	0.633	0.344	0.366	0.849	0.291	0.594
BACE	GCN	IG	scaffold	200	0.628	0.698	0.116	0.807	0.138	0.567	0.352	0.352	0.865	0.227	0.668
BACE	GINE	IG	random	200	0.730	0.821	0.062	0.788	0.116	0.557	0.274	0.268	0.691	0.126	0.323
BACE	GINE	IG	scaffold	200	0.545	0.716	0.061	0.863	0.093	0.297	0.239	0.271	0.564	0.415	0.174
BBBP	AttentiveFP	IG	random	200	0.757	0.884	0.038	0.749	0.234	0.418	0.092	0.093	0.853	0.200	0.104
BBBP	AttentiveFP	IG	scaffold	200	0.855	0.954	0.127	0.750	0.409	0.562	0.251	0.169	0.847	0.327	0.070
BBBP	GAT	IG	random	200	0.735	0.900	0.032	0.770	0.002	0.447	0.219	0.219	0.876	0.193	0.291
BBBP	GAT	IG	scaffold	200	0.812	0.914	0.123	0.788	0.123	0.522	0.262	0.322	0.941	0.158	0.444
BBBP	GCN	IG	random	200	0.740	0.889	0.043	0.796	−0.143	0.425	0.155	0.157	0.842	0.092	0.195
BBBP	GCN	IG	scaffold	200	0.823	0.915	0.115	0.801	0.189	0.528	0.302	0.357	0.893	0.080	0.506
BBBP	GINE	GNNExpl.	random	200	0.730	0.892	0.033	0.799	−0.018	0.397	0.183	0.216	0.955	0.241	0.164
BBBP	GINE	GNNExpl.	scaffold	200	0.827	0.925	0.115	0.804	0.349	0.487	0.330	0.355	0.934	0.327	0.400
BBBP	GINE	IG	random	200	0.730	0.892	0.033	0.770	0.044	0.427	0.219	0.220	0.890	0.206	0.314
BBBP	GINE	IG	scaffold	200	0.827	0.925	0.115	0.763	**0.509**	0.500	0.282	0.363	0.855	0.277	0.471
BBBP	GINE	PGExpl.	random	200	0.730	0.892	0.033	0.899	−0.131	0.285	0.126	0.202	0.370	0.159	—
BBBP	GINE	PGExpl.	scaffold	200	0.827	0.925	0.115	0.882	0.212	0.257	0.189	0.331	0.440	0.222	—
BBBP	MPNN	IG	random	200	0.757	0.906	0.040	0.792	0.047	0.373	0.109	0.169	0.860	0.113	0.142
BBBP	MPNN	IG	scaffold	200	0.818	0.923	0.115	0.819	0.160	0.502	0.175	0.323	0.882	0.080	0.307

Table 8: Molecular classification cells (no node-level ground truth available), 2 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets, since no per-atom labels are published, so only model-side reliability can be measured, which is the gap the Tier-1 cells are needed to close. *charact.* is the GraphFramEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules; both are blank for carried rows (^{c}), whose per-molecule records this run did not regenerate. Bold marks the highest occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid−	stab.	charact.	unfaith.
ClinTox	GINE	GNNExpl.	random	200	0.730	0.880	0.101	0.476	**0.275**	0.433	0.215	0.174	0.975	0.129	0.340
ClinTox	GINE	GNNExpl.	scaffold	200	0.673	0.844	0.144	0.670	−0.027	0.437	0.237	0.222	0.988	0.129	0.319
ClinTox	GINE	IG	random	200	0.730	0.880	0.101	0.799	−0.287	0.390	0.177	0.174	0.911	0.129	0.205
ClinTox	GINE	IG	scaffold	200	0.673	0.844	0.144	0.756	−0.170	0.417	0.222	0.222	0.971	0.129	0.221
DILI	GINE	IG	random	142	0.714	0.761	0.117	0.742	0.299	0.472	0.192	0.209	0.813	0.220	0.287
DILI	GINE	IG	scaffold	142	0.737	0.801	0.085	0.826	**0.336**	0.582	0.237	0.261	0.921	0.282	0.527
SIDER	GCN	IG	random	200	0.643	0.660	0.061	0.744	0.159	0.568	0.100	0.120	0.750	0.154	0.054
SIDER	GCN	IG	scaffold	200	0.575	0.668	0.028	0.748	0.365	0.722	0.181	0.312	0.879	0.179	0.033
SIDER	GINE	IG	random	200	0.642	0.671	0.070	0.764	0.277	0.613	0.174	0.241	0.869	0.243	0.449
SIDER	GINE	IG	scaffold	200	0.555	0.627	0.033	0.753	**0.436**	0.655	0.233	0.287	0.866	0.293	0.511
Tox21	GINE	IG	random	200	0.940	0.820	0.113	0.749	**−0.269**	0.333	−0.048	−0.042	0.710	0.037	0.309
Tox21	GINE	IG	scaffold	200	0.942	0.745	0.040	0.794	−0.413	0.402	−0.070	−0.068	0.927	0.042	0.149
hERG	GINE	IG	random	197	0.655	0.821	0.109	0.761	0.240	0.624	0.384	0.384	0.953	0.188	0.809
hERG	GINE	IG	scaffold	197	0.472	0.653	0.468	0.791	**0.743**	0.877	0.730	0.730	0.943	0.074	0.642

References

[1] M. Sundararajan, A. Taly, and Q. Yan. Axiomatic attribution for deep networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR 70:3319–3328, 2017.

[2] K. Simonyan, A. Vedaldi, and A. Zisserman. Deep inside convolutional networks: visualising image classification models and saliency maps. *International Conference on Learning Representations (ICLR) Workshop*, 2014.

[3] J. T. Springenberg, A. Dosovitskiy, T. Brox, and M. Riedmiller. Striving for simplicity: the all convolutional net. *International Conference on Learning Representations (ICLR) Workshop*, 2015.

[4] Z. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec. GNNExplainer: generating explanations for graph neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[5] D. Luo, W. Cheng, D. Xu, W. Yu, B. Zong, H. Chen, and X. Zhang. Parameterized explainer for graph neural network. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[6] H. Yuan, H. Yu, J. Wang, K. Li, and S. Ji. On explainability of graph neural networks via subgraph explorations. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, PMLR 139:12241–12252, 2021.

[7] P. E. Pope, S. Kolouri, M. Rostami, C. E. Martin, and H. Hoffmann. Explainability methods for graph convolutional neural networks. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[8] H. Yuan, H. Yu, S. Gui, and S. Ji. Explainability in graph neural networks: a taxonomic survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(5):5782–5799, 2023.

[9] C. Agarwal, O. Queen, H. Lakkaraju, and M. Zitnik. Evaluating explainability for graph neural networks. *Scientific Data*, 10:144, 2023.

[10] K. Amara, Z. Ying, Z. Zhang, Z. Han, Y. Zhao, Y. Shan, U. Brandes, S. Schemm, and C. Zhang. GraphFramEx: towards systematic evaluation of explainability methods for graph neural networks. *Proceedings of the First Learning on Graphs Conference (LoG)*, PMLR 198, 2022.

[11] M. Liu, Y. Luo, L. Wang, Y. Xie, H. Yuan, S. Gui, H. Yu, Z. Xu, J. Zhang, Y. Liu, K. Yan, H. Liu, C. Fu, B. M. Oztekin, X. Zhang, and S. Ji. DIG: a turnkey library for diving into graph deep learning research. *Journal of Machine Learning Research*, 22(240):1–9, 2021.

[12] M. Hiltscher, M. Bianciotto, and F. Grisoni. Explaining What Matters: Faithfulness in Molecular Deep Learning. *ChemRxiv*, 2026. doi:10.26434/chemrxiv.10001594/v1. Platform: `https://github.com/molML/MolFaith`.

[13] L. Faber, A. K. Moghaddam, and R. Wattenhofer. When comparing to ground truth is wrong: on evaluating GNN explanation methods. *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pages 332–341, 2021.

[14] B. Sanchez-Lengeling, J. N. Wei, B. K. Lee, E. Reif, P. Wang, W. W. Qian, K. McCloskey, L. J. Colwell, and A. B. Wiltschko. Evaluating attribution for graph neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[15] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity checks for saliency maps. *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[16] S. Hooker, D. Erhan, P.-J. Kindermans, and B. Kim. A benchmark for interpretability methods in deep neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[17] N. Kokhlikyan, V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, and O. Reblitz-Richardson. Captum: a unified and generic model interpretability library for PyTorch. *arXiv:2009.07896*, 2020.

[18] M. Fey and J. E. Lenssen. Fast graph representation learning with PyTorch Geometric. *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.

[19] G. Landrum et al. RDKit: open-source cheminformatics. `https://www.rdkit.org`.

[20] K. Xu, W. Hu, J. Leskovec, and S. Jegelka. How powerful are graph neural networks? *International Conference on Learning Representations (ICLR)*, 2019.

[21] W. Hu, B. Liu, J. Gomes, M. Zitnik, P. Liang, V. Pande, and J. Leskovec. Strategies for pre-training graph neural networks. *International Conference on Learning Representations (ICLR)*, 2020.

[22] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations (ICLR)*, 2017.

[23] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio. Graph attention networks. *International Conference on Learning Representations (ICLR)*, 2018.

[24] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl. Neural message passing for quantum chemistry. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR 70:1263–1272, 2017.

[25] Z. Xiong, D. Wang, X. Liu, F. Zhong, X. Wan, X. Li, Z. Li, X. Luo, K. Chen, H. Jiang, and M. Zheng. Pushing the boundaries of molecular representation for drug discovery with the graph attention mechanism. *Journal of Medicinal Chemistry*, 63(16):8749–8760, 2020.

[26] K. Iyer and N. Sabar. M-JEPA: predictive self-supervised learning for molecular graphs with scaffold-shift evaluation on Tox21. *Journal of Chemical Information and Modeling*, 66(14):8008–8021, 2026. doi:10.1021/acs.jcim.6c00828.

[27] Z. Wu, B. Ramsundar, E. N. Feinberg, J. Gomes, C. Geniesse, A. S. Pappu, K. Leswing, and V. Pande. MoleculeNet: a benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018.

[28] K. Huang, T. Fu, W. Gao, Y. Zhao, Y. Roohani, J. Leskovec, C. W. Coley, C. Xiao, J. Sun, and M. Zitnik. Therapeutics Data Commons: machine learning datasets and tasks for drug discovery and development. *NeurIPS Datasets and Benchmarks Track*, 2021.

[29] A. K. Debnath, R. L. Lopez de Compadre, G. Debnath, A. J. Shusterman, and C. Hansch. Structure-activity relationship of mutagenic aromatic and heteroaromatic nitro compounds. Correlation with molecular orbital energies and hydrophobicity. *Journal of Medicinal Chemistry*, 34(2):786–797, 1991.

[30] C. Morris, N. M. Kriege, F. Bause, K. Kersting, P. Mutzel, and M. Neumann. TUDataset: a collection of benchmark datasets for learning with graphs. *ICML Workshop on Graph Representation Learning and Beyond*, 2020.

[31] G. W. Bemis and M. A. Murcko. The properties of known drugs. 1. Molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996.

[32] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR 70:1321–1330, 2017.